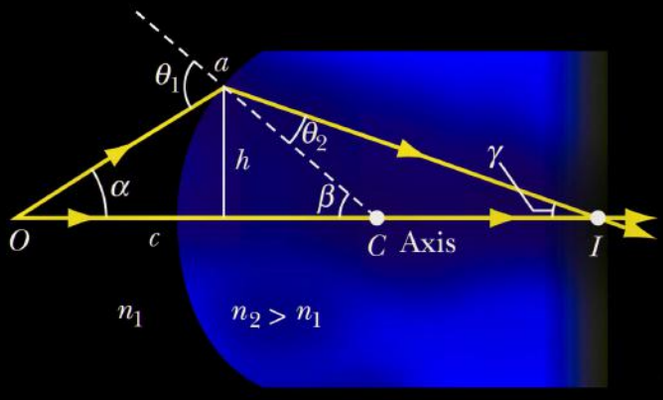
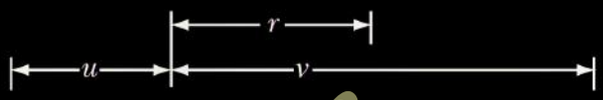
Refraction from Spherical Surfaces:

Just like with mirrors, a sharp image formation from spherical refracting surface requires paraxial rays. Let O be an object and r be the radius of curvature of the spherical surface. Now using snell's law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$ (1)

$\Rightarrow n_1\theta_1 \approx n_2\theta_2$





Also: $\alpha + \beta = \theta_1$ (2)

and $\beta - \gamma = \theta_2$ (3)

$$\Rightarrow \frac{\alpha+\beta}{\beta-\gamma} = \frac{\theta_1}{\theta_2} = \frac{n_2}{n_1} \quad \text{(using (1))} \quad (4)$$

Now for small angles, $\tan x \approx x$ thus $\alpha \approx \tan\alpha = h/-u$

$\beta \approx h/r$, $\gamma = h/v$ thus using in (4):

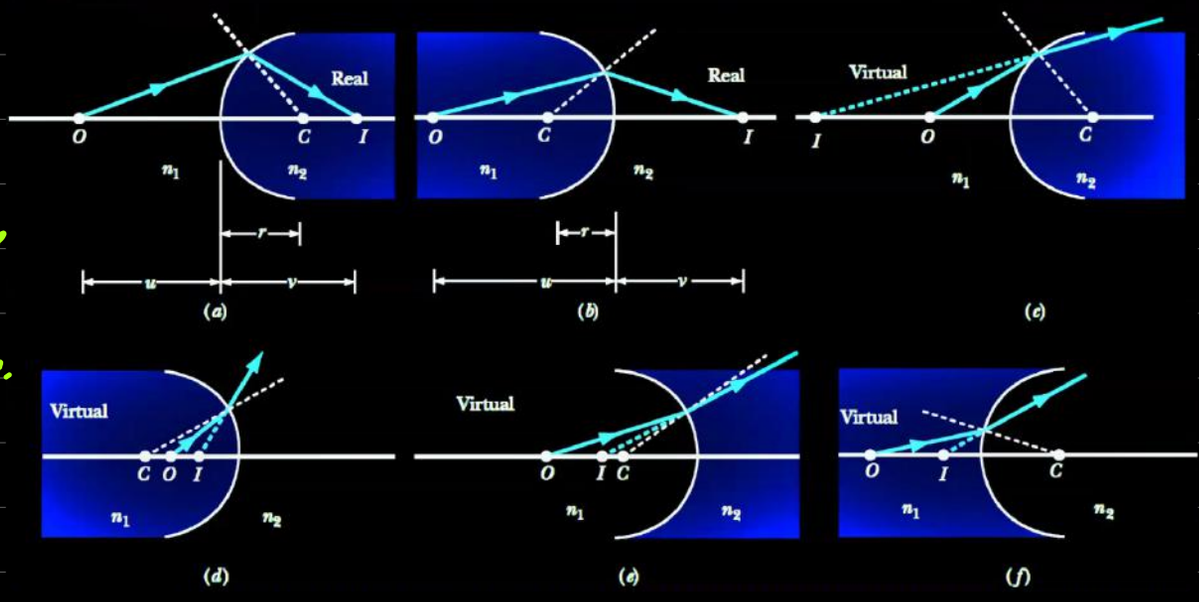
$$\frac{-h/u + h/r}{h/r - h/v} = n_2/n_1$$ rearranging after canceling h:

$$\boxed{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{r}}$$

→ Here u, v and r are all signed values

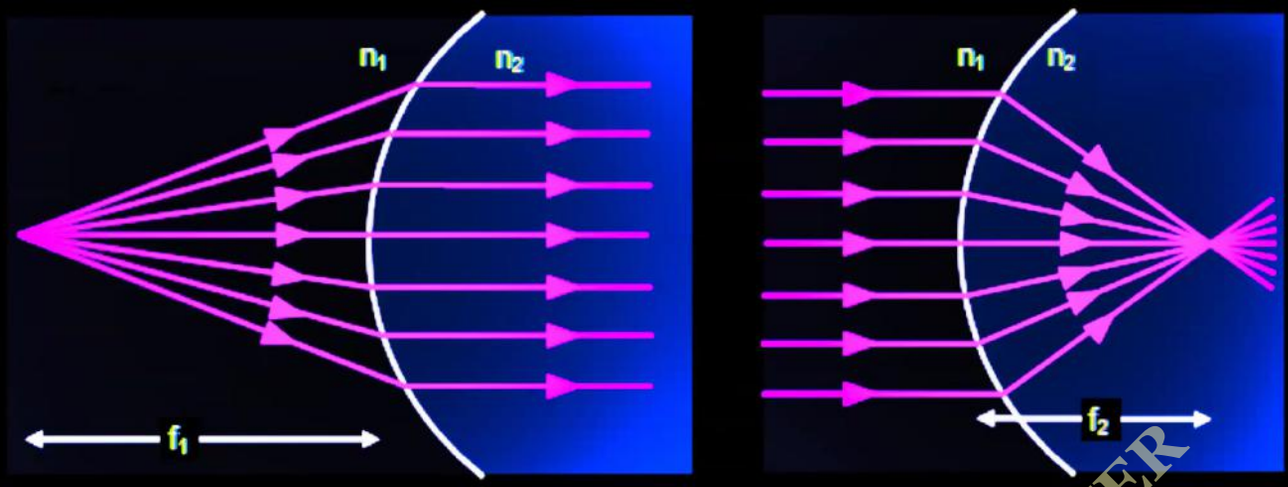
→ 1 is the medium of incident ray and 2 is the medium of refracted ray.

→ Image can be real or virtual depending on object distance and nature of refracting surfaces.



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Foci Of Spherical Surfaces:



With spherical surfaces we have two distinct foci defined as:

(a) f_1 : Object coordinate for which image is formed at ∞
(b) f_2 : Image coordinate when the object is at infinity.

Now for f_1, $u = f_1$, $v = \infty$ thus using spherical surface formula:

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R} \Rightarrow \frac{-n_1}{f_1} = \left(\frac{n_2 - n_1}{R}\right) \quad (1)$$

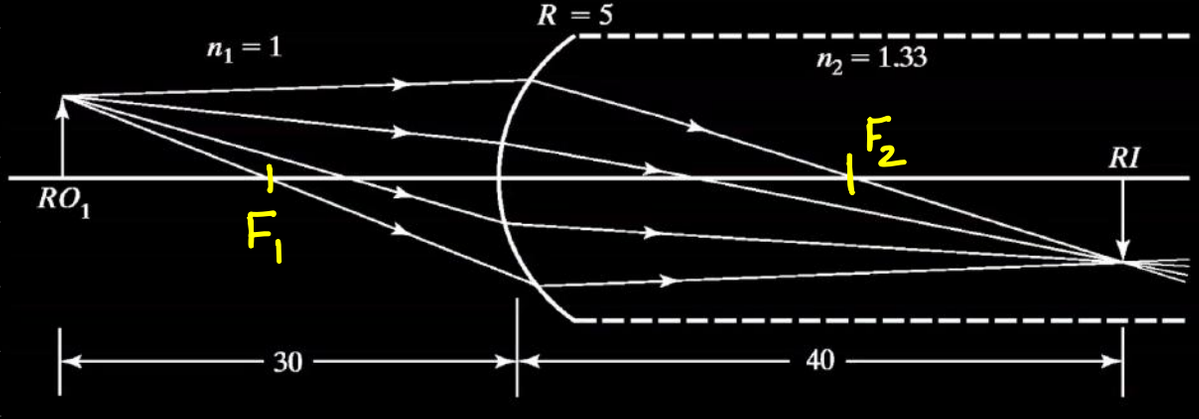
Similarly for f_2, $u = -\infty$, $v = f_2$ thus:

$$\frac{n_2}{f_2} = \left(\frac{n_2 - n_1}{R}\right) \quad (2)$$

From (1) & (2): $\boxed{\frac{f_1}{f_2} = -\frac{n_1}{n_2}}$ Also $\boxed{\frac{f_2}{v} + \frac{f_1}{u} = 1}$

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Ray Diagram For Image Formation From Spherical Surfaces:



For ray tracing from spherical surfaces, we typically use following two rays:

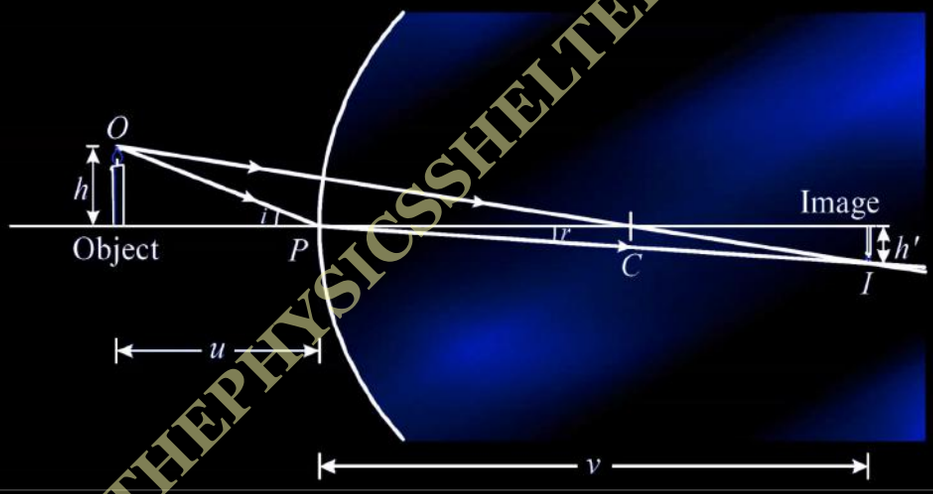
① A ray incident parallel to principal axis passes through f_2

② A ray incident through f_1 goes parallel to principal axis after refraction.

→ In Newtonian convention we can prove that $xy = f_1 f_2$ where x is the object coordinate measured from f_1 and y is image coordinate as measured from f_2.

TRANSVERSE MAGNIFICATION FROM SPHERICAL SURFACES:

Consider adjacent ray diagram:



magnification:

$$m = \frac{-h'}{h} = \frac{-v \tan r}{-u \tan i}$$

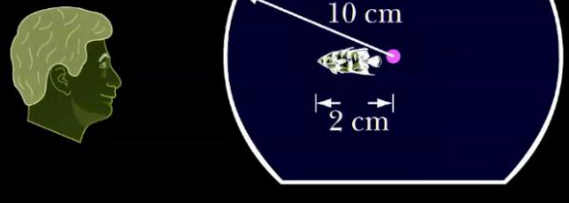
Now by snell's law:

$$n_1 \tan i \approx n_2 \tan r$$

$$\Rightarrow \boxed{m = \frac{n_1 v}{n_2 u}}$$

EXAMPLE:

Assume that a boy views a fish in a spherical aquarium of radius 10 cm kept at a distance of 2 cm from the center toward the left of center. The boy is viewing from the left (See Figure). Where will the image of fish be formed?



Solution: $n_1 = 1.33$ $n_2 = 1$, $u = -8$, $R = -10$, $v = ?$

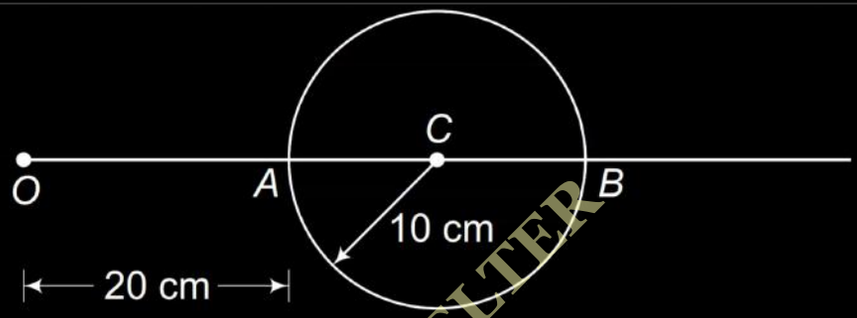
$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \Rightarrow \frac{1}{v} - \frac{4}{3 \times (-8)} = \left(\frac{1 - 4/3}{-10}\right)$$

$$\Rightarrow \frac{1}{v} = \frac{-1}{6} + \frac{1}{30} = \frac{-4}{30} \Rightarrow v = -7.5 \text{ cm}.$$ Thus image is formed 7.5 cm to the right of the aquarium's refracting wall.

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Example:

A glass sphere of radius R = 10 cm is kept inside water. A point object O is placed at 20 cm from A as shown in figure. Find the position and nature of the image when seen from other side of the sphere. Also draw the ray diagram. Given, $\mu_g = 3/2$ and $\mu_w = 4/3$.



Solution: For refraction at A,

$\mu_1 = \frac{4}{3}$, $\mu_2 = 3/2$, $u = -20$

$R = +10$ cm

Now using sphere formula:

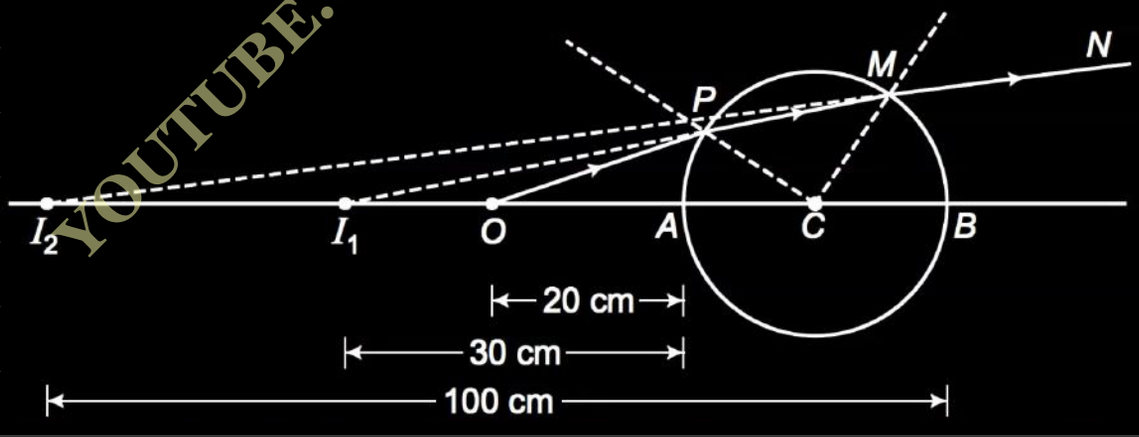
$$\frac{n_2}{v} - \frac{n_1}{u} = \left(\frac{n_2 - n_1}{R}\right) \Rightarrow \frac{3/2}{v} - \frac{4/3}{-20} = \left(\frac{3/2 - 4/3}{10}\right) \Rightarrow v = -30 \text{ cm.}$$

Now for second surface, $u = -30 - 20 = -50$, $\mu_2 = 4/3$, $\mu_1 = 3/2$, $R = -10$ thus:

$$\frac{4/3}{v} - \frac{3/2}{-50} = \frac{4/3 - 3/2}{-10} \Rightarrow v = -100 \text{ cm,}$$

$\Rightarrow$ final image is formed 100 cm to the left of B. Ans.

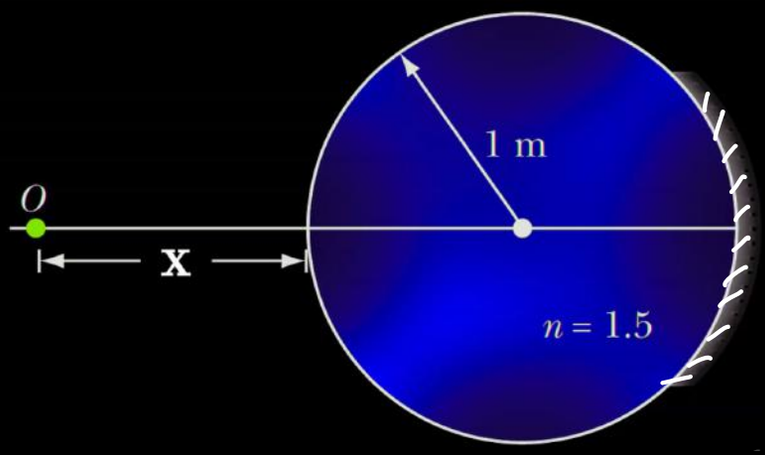
The ray diagram is as shown in the following figure.



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EXAMPLE:

Consider a silvered sphere of radius 1m as shown in adjacent figure. Find the possible values of x for Auto-collimation.



Solution: Just as we studied in the chapter on reflection, here also, for autocollimation we have either ray reversal (intermediate image at centre of curvature) or ray exchange i.e intermediate image at pole. For image at centre of curvature,

$$\frac{\mu}{R} - \frac{1}{-x} = \frac{(\mu-1)}{R} \Rightarrow \frac{1}{x} = \frac{-1}{R} \Rightarrow x = -R,$$ however this would imply a virtual object (since object is across the refracting boundary. This simply means a convergent pencil falling on sphere with every ray normal to the surface of the sphere.

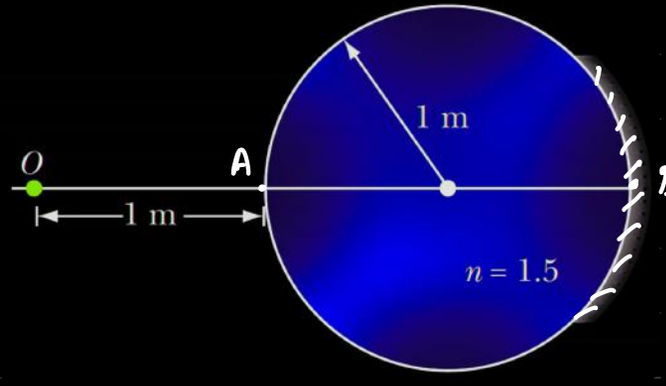
Now for image at pole:

$$\frac{\mu}{2R} - \frac{1}{-x} = \frac{(\mu-1)}{R} \Rightarrow \frac{1}{x} = \left(\frac{\mu-2}{2R}\right)$$

$$\Rightarrow x = \frac{2R}{\mu-2} = \frac{2}{-0.5} = -4\,m,$$ in this case also we have virtual object. Thus auto collimation is possible for two positions of virtual object i.e $x = +1\,m$ or $x = +4\,m$ Ans.

EXAMPLE:

Locate the final image of the object shown in the figure. (Silvered at B)



Solution: Here there are three processing events: Refraction at A then Reflection at B and again Refraction at A.

Now for first event:

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

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$$\Rightarrow \quad \frac{3}{2v} - \frac{1}{-1} = \left(\frac{3}{2} - 1\right)\cdot\frac{1}{1} \quad \Rightarrow \quad \frac{3}{2v} = -\frac{1}{2} \quad \Rightarrow \quad v = -3\ m$$

Now for the mirror, $u = -3 - 2 = -5\ m$, $f = \frac{R}{2} = -1/2$

Now $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-5} = \frac{1}{-1/2} = -2 \Rightarrow v = -\frac{5}{9}\ m$

i.e $\frac{5}{9}$ m to the left of B. Now incident rays reverse their direction and we must apply the sign convention very carefully.

Now for second refraction at A, $u = -\left[2 - \frac{5}{9}\right] = -13/9$ m.

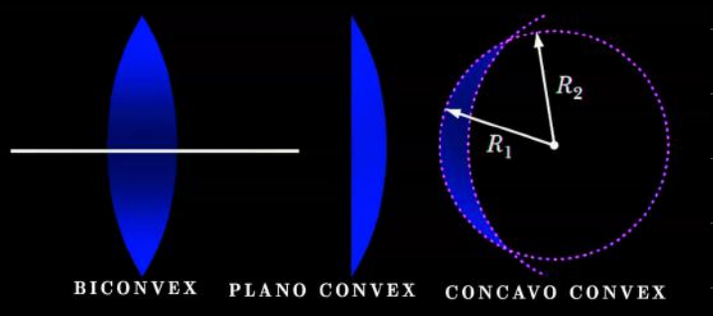
$\mu_2 = 1$, $\mu_1 = 3/2$, $R = -1$ thus: $\frac{1}{v} - \frac{3/2}{(-13/9)} = \left(\frac{1 - 3/2}{-1}\right) = \frac{1}{2}$

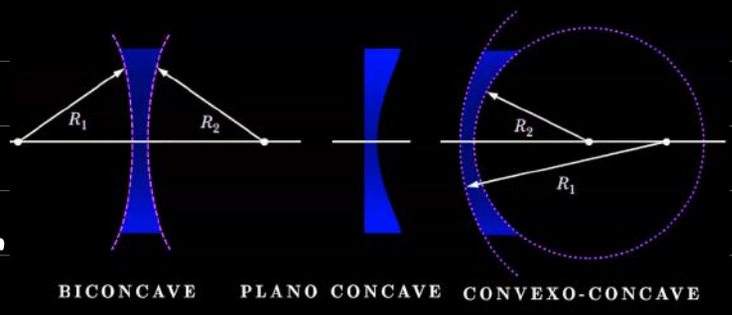
$\Rightarrow \quad v = -13/7$ m. Thus final image is formed at 13/7 m to the right of A. Ans.

→ Negative v means final image is virtual and formed inside the sphere.

Nomenclature Of Thin Lenses:

While naming the lenses, second half of the name signifies the nature of surface with smaller magnitude of radius of curvature. Thus a concavo convex lens has a smaller radius of curvature on convex side than on the concave side. The name readily tells us whether a lens is converging or diverging when used in a medium of refractive index smaller than refractive index of the material of lens.





Thus any lens with word "convex" at the end is a converging lens when placed in air and the one with word "concave"

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at the end is a diverging lens when placed in air (or a medium of smaller refractive index than the lens itself). When placed in a medium of refractive index more than the material of the lens itself, the converging / diverging nature toggles.

→ A concavo convex lens is thicker in the middle and thinner at the periphery

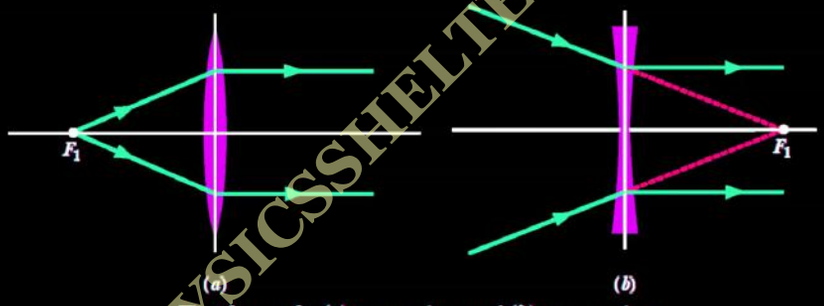
→ A convexo concave lens is thinner in the middle and thicker at the periphery.

First Focus (Object Focus) And Second Focus (Image Focus) Of A Lens:

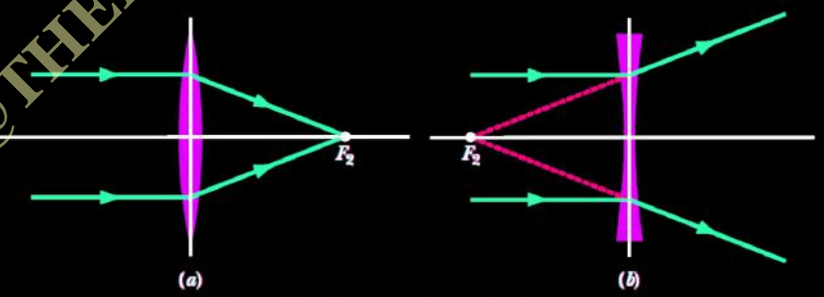
First Focus: location of object for which image is formed at infinity is called first focus or object focus.

Second Focus (Image Focus): The location of image for an object placed at infinity is called second focus or image focus.

The two foci for concave and convex lenses are shown in the adjacent figure.



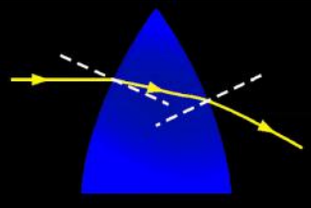
First focus of a (a) convex lens and (b) concave lens.



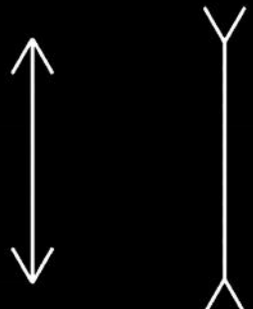
Second focus of a (a) convex lens and (b) concave lens.

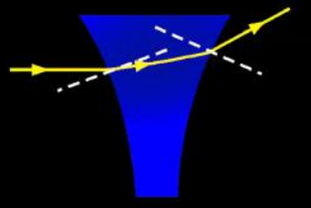
→ Normally focal length of a lens refers to the image focus.

→ Although for simplicity, while drawing the ray diagrams, we refract the ray at mid-plane of the lens, infact, refraction is occuring at both surfaces.



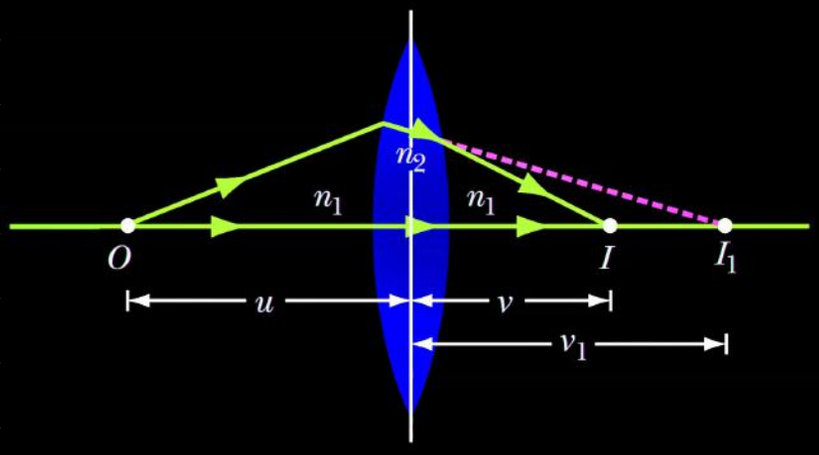
→ Sometimes in ray diagrams we omit the lenses and show them by symbols as shown in adjacent figure.





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Derivation Of Lens Maker's Formula:



Consider the ray diagram for the lens shown in adjacent figure.

For the first interface:

$$\frac{n_2}{v_1} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R_1} \quad (1)$$

Here v_1 is intermediate image coordinate. Now since the lens is thin, we can also treat v_1 as object coordinate for second refraction.

Thus for second refraction:

$$\frac{n_1}{v} - \frac{n_2}{v_1} = \frac{(n_1 - n_2)}{R_2} \quad (2)$$

Now adding (1) and (2):

$$n_1\left[\frac{1}{v} - \frac{1}{u}\right] = (n_2 - n_1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] \quad \text{or}$$

$$\frac{1}{v} - \frac{1}{u} = \left[\frac{n_2}{n_1} - 1\right]\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$, now by definition of focal length,

when $u = -\infty$, $v = f$ thus $\boxed{\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]}$

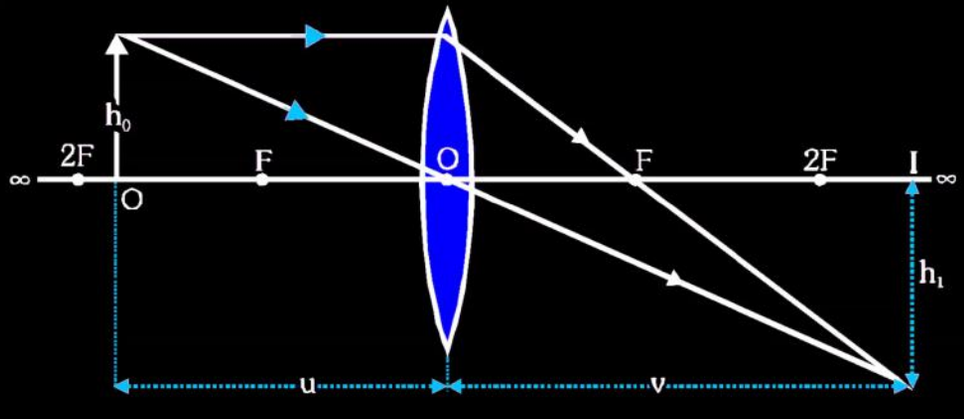
→ f, R_1 and R_2 are all signed values according to Cartesian Convention.

→ Now using the definition of focal length, our lens equation simplifies to:

$$\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}}$$

The f refers to the second focus here.

Image Formation Rules For Converging Lens:

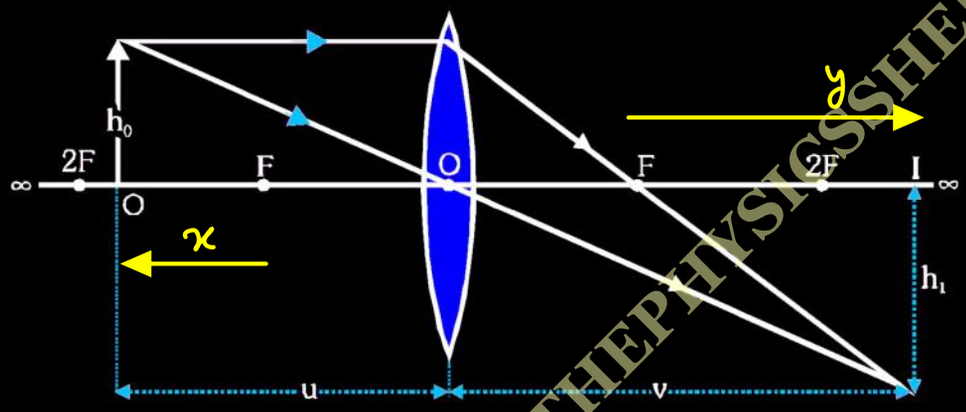


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Object	Image	Magnification
$-\infty$	F	$\|m\| \ll 1$ & $m<0$
$-\infty - 2F$	F–2F	$\|m\| < 1$ & $m<0$
2F	2F	$m = -1$
F – 2F	∞–2F	$\|m\| > 1$ & $m<0$
Just before F	$+\infty$	$m \ll -1$
Just after F towards O	$-\infty$	$m \gg 1$
F – O	In front of lens	$m > 1$

Lens Formula In Newtonian Convention:



In Newtonian convention, coordinates of object and image are measured from the respective foci (see x & y above). Thus $v = f + y$ and $u = -f + x$ (x & y are signed values, f is +ve for converging lenses). Thus:

$$\frac{1}{f+y} - \frac{1}{-f+x} = \frac{1}{f} \Rightarrow \boxed{xy = -f^2}$$

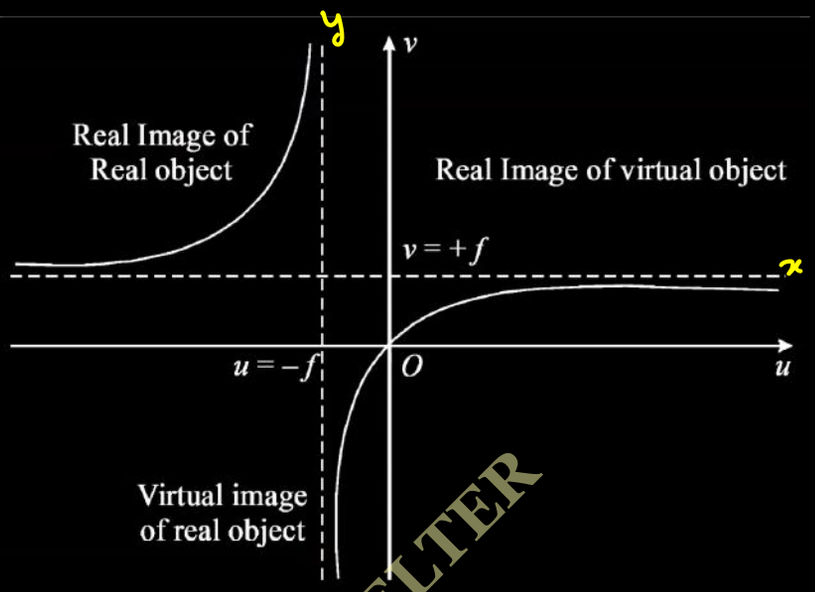
Thus object and image are always on the opposite side of their respective foci, (∵ when $x>0$, $y<0$).

→ Students should either avoid using this formula for diverging lenses or use it with great caution, there is a high probability of making a silly mistake.

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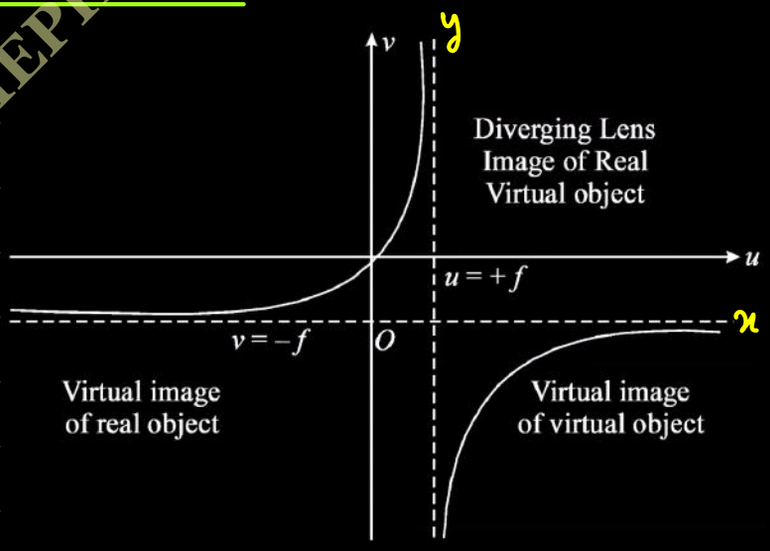
x-y And u-v Graphs For Convex Lens:

Recall from previous section that $xy = -f^2$ or $(u+f)(v-f) = -f^2$ thus if origin were to maintain its place, uv graph would be shifted f units to the left and f units upwards. However instead of shifting the graph, we are shifting the origin thus, origin will be shifted f units rightwards and f units downwards. This gives us the above graph for the convex lens in variables x-y and u-v.



x-y And u-v Graphs For Diverging Lens:

For a diverging lens also $xy = -f^2$ holds however f is negative. This can once again be written as $(u+f)(v-f) = -f^2$ thus the origin needs to be shifted f units to the right and f units down, but $\because f$ is negative, the actual shifting is left and up.

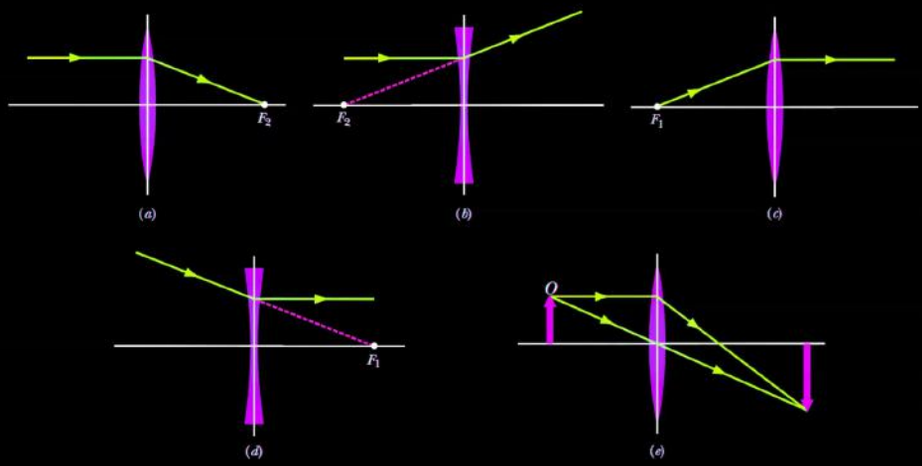


Convenient Rays For Ray Tracing For Lenses:

① Ray ∥ to principal axis passes through F_2 after refraction

② A Ray passing through F_1 becomes parallel to principal axis after Refraction

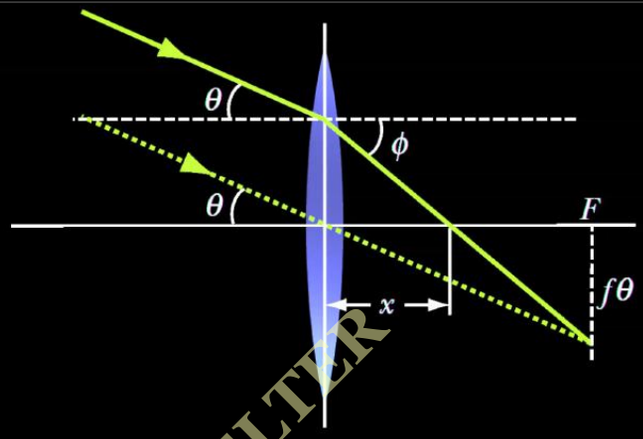
③ A ray through optical centre goes undeviated.



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Focal Plane: A plane perpendicular to the principal axis passing through the focus is called the focal plane. If a paraxial beam of parallel rays is incident on a converging lens, it gets focused in the focal plane instead of the focus.

→ If the beam makes an angle θ with principal axis, it is focused at a distance $f\theta$ below the principal axis (see figure). This ray diagram can be used to find angle of deviation of a ray incident at a height h above the optical centre.



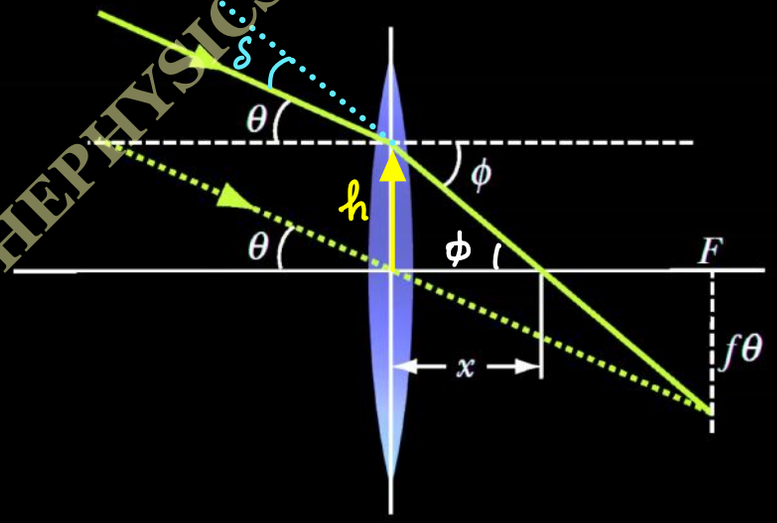
Angle Of Deviation Of A Ray Incident At Height h Above Principal Axis:

Consider the adjacent ray diagram. Here by similar triangles:

$$\frac{h}{x} = \frac{f\theta}{f-x} = \phi$$

solving we get

$$x = \frac{hf}{h+f\theta}, \quad \phi = \frac{h+f\theta}{f}$$



Now angle of deviation is $\phi - \theta$

i.e $\delta = \phi - \theta = h/f$ thus angle of deviation of a ray incident at a height h above optical centre is given as:

$$\boxed{\delta = h/f}$$

→ **Alternative derivation:** Imagine a ray parallel to principal axis at a height h above optical centre. This ray must pass through the focus thus have an angle of deviation given as $\tan\delta \approx \delta = h/f$. Now since lens can be thought of as made of several pieces of thin prisms, and for thin prism angle of deviation for near normal incidence is $\delta = (\mu-1)A$ irrespective of incidence angle thus true also for every ray, $\delta = h/f$.

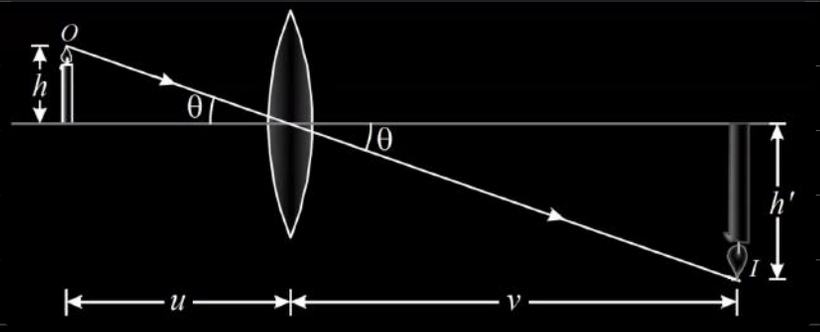
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Magnification From Thin Lens: Transverse (Lateral Magnification)

Consider the ray diagram shown in adjacent figure. The transverse magnification is given by

$$m = -\frac{h'}{h} = -\frac{v\tan\theta}{-u\tan\theta}$$

thus $\boxed{m = v/u}$



Transverse Magnification In Newtonian Convention:

According to newtonian convention, $v = f + y$ and $u = -f + x$

thus,

$$m = \frac{f+y}{x-f} \qquad \text{but } xy = -f^2 \text{ thus}$$

$$m = \frac{f - f^2/x}{x-f} = \frac{1}{x}\left[\frac{f(x-f)}{(x-f)}\right] = f/x$$

Thus $\boxed{m = f/x = -y/f}$

→ This means that for equidistant object locations from the object focus, magnitude of magnification is same although the sign is inverted.

Longitudinal Magnification From A Thin Lens:

Suppose a very small object lies along axis of a lens extending between coordinates u and $u+du$ and let its image extend between coordinates v and $v+dv$ then its longitudinal magnification is given by $m_L = dv/du$

Now $$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow -\frac{1}{v^2}\frac{dv}{du} + \frac{1}{u^2} = 0 \Rightarrow \frac{dv}{du} = v^2/u^2$$

Now recall that transverse magnification, $m = v/u$ thus longitudinal magnification can be written as:

$$\boxed{m_L = m^2 = \frac{v^2}{u^2} = f^2/x^2 = y^2/f^2}$$

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→ Since longitudinal magnification is perfect square, the image is never longitudinally inverted.

Relation Between Axial Object Velocity And Image Velocity:

Using lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{-1}{v^2}\frac{dv}{dt} + \frac{1}{u^2}\frac{du}{dt} = 0$

$\Rightarrow \frac{dv}{dt} = \frac{v^2}{u^2}\frac{du}{dt}$ thus $\boxed{v_{I_A} = m_L^2 v_{O_A}}$

(v_{I_A} → Axial velocity of image, v_{O_A} : Axial velocity of object).

Relation Between Transverse Object Velocity And Image Velocity:

Let h_o be the transverse coordinate of object tip and h_i be the transverse coordinate of image tip then:

$$h_i = \frac{v}{u} h_o = \frac{f}{x} h_o \Rightarrow \frac{dh_i}{dt} = f\left[h_o\left(\frac{-1}{x^2}\frac{dx}{dt}\right) + \frac{1}{x}\frac{dh_o}{dt}\right]$$

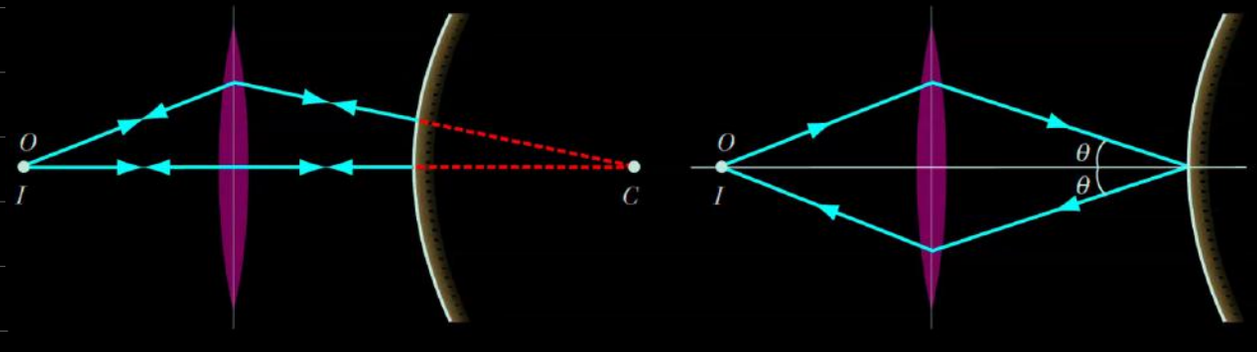
$$\Rightarrow \frac{dh_i}{dt} = \frac{f}{x}\left[v_{o_T} - \frac{h_o}{x} v_{o_A}\right]$$

$$\Rightarrow v_{i_T} = m\left[v_{oT} - \frac{h_o}{x} v_{oA}\right]$$ v_{i_T} : Transverse velocity of image

v_{oT} : Transverse velocity of object, v_{oA} : Axial velocity of object.

Example: Find the distance between a convex lens and a convex mirror both of focal length 20 cm so that for an object placed 30 cm in front of the lens, the final image is at the object itself.

Solution:



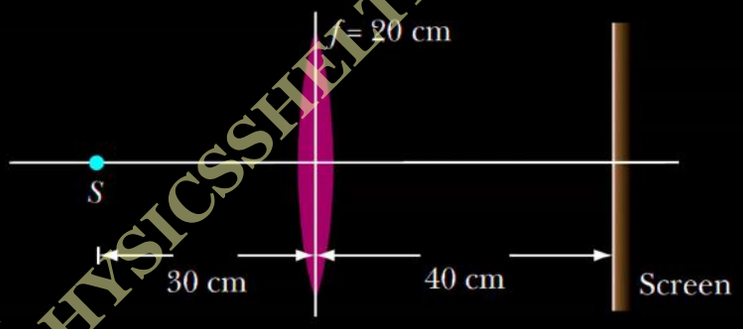
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Recall that we can achieve autocollimation in two ways i.e ray reversal or ray exchange. In one case intermediate image must be formed at centre of curvature, while in second case, it must be formed at the pole of the mirror. Thus applying lens formula:

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{-30} = \frac{1}{20} \Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60}$$

thus $v = 60$ cm. Now for mirror, $R = 2f = 40$ cm. Thus two possible locations of mirror: Case I: $d = 60$ cm Case II, $d = 60 - 40 = 20$ cm Ans.

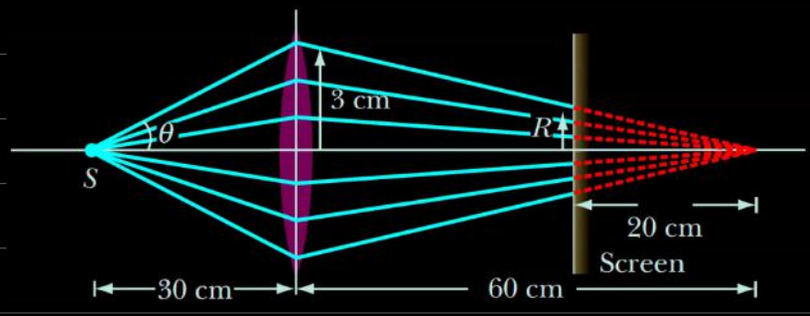
EXAMPLE: Find the radius of the light spot on the screen as shown in the figure. Consider only the rays refracted from the lens. Find the average intensity of the spot. The source is isotropic emitting a total power 100 W. The aperture of lens is 6cm.



Solution: Applying lens formula:

$$\frac{1}{v} - \frac{1}{-30} = \frac{1}{20} \Rightarrow v = 60\text{cm}.$$

This gives us the ray diagram as shown in the figure. Now, By similar triangles:



$$\frac{R}{20} = \frac{3}{60} \Rightarrow R = 1\text{cm}$$ Ans.

Now Power collected by lens is proportional to its solid angle subtended on the source and for intensity we need to divide this power by area of the spot. Thus, required intensity:

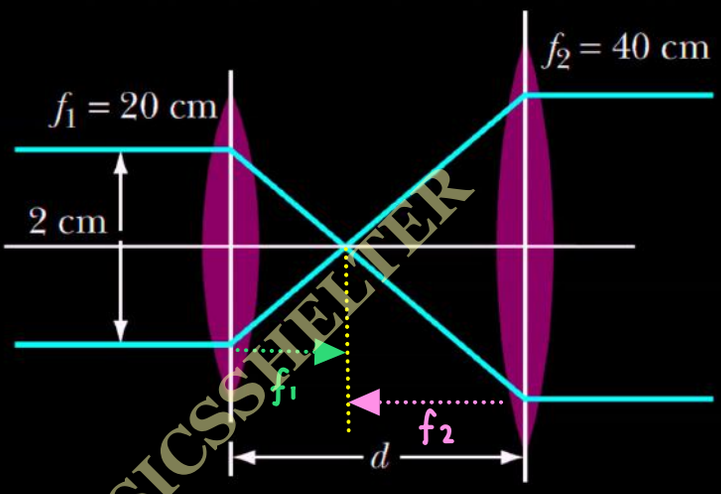
$$I = P \times \frac{2\pi(1-\cos\theta)}{4\pi} \cdot \frac{1}{\pi R^2}, \text{ Now for small } \theta,$$

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$$1-\cos\theta = 2\sin^2(\theta/2) \approx 2\tan^2(\theta/2) = \frac{1}{2}\theta^2 = \frac{1}{2}\times\left(\frac{3}{30}\right)^2 = \frac{1}{200}$$

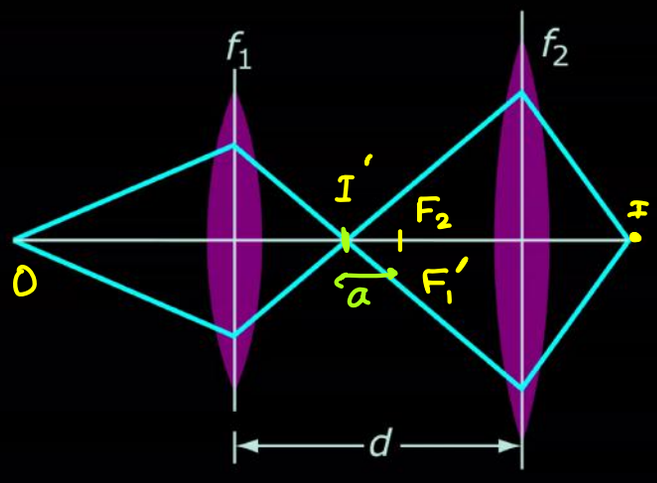
Thus required intensity: $I = \dfrac{100}{2\pi\times(0.01)^2}\times\dfrac{1}{200} = \dfrac{2500}{\pi}$ W/m² Ans.

Beam Expander / Contractor Doublet:



Consider the doublet ray diagram shown in adjacent figure. If two lenses of focal length f_1 and f_2 are separated by a distance $d = f_1 + f_2$, then such a combination acts as a beam expander or contractor with aperture ratio of final beam to initial beam being f_2/f_1 (as can be readily seen from similar triangles). If $f_1 > f_2$ the combination acts as a beam expander while if $f_2 < f_1$, the combination acts like a beam contractor.

Constant Magnification Doublet:



If two lenses are separated by a distance $f_1 + f_2$ where f_1 and f_2 are individual focal lengths, the pair serves as a system whose total transverse magnification is $\dfrac{f_2}{f_1}$ irrespective of the object position. This can be readily seen as follows.

Let a be the distance of intermediate image from the common focal point of the two lenses (F_2, F_1'). Now recall that magnification from a lens is given as $m = \dfrac{f}{x} = -y/f$ thus for the first lens, magnification, $m_1 = \dfrac{a}{f_1}$ and for second lens, magnification,

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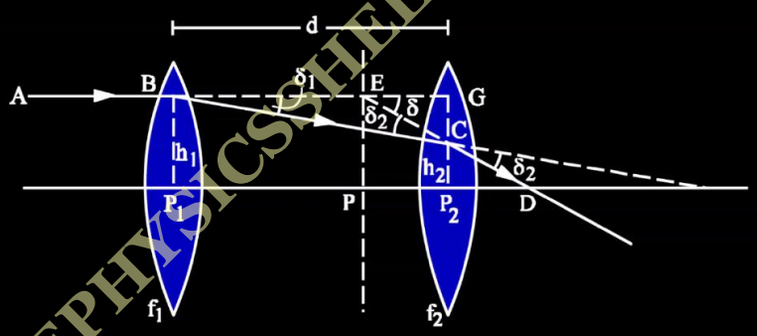
$m_2 = \frac{f_2}{-a}$ thus total magnification, $m = m_1 \times m_2 = \frac{a}{f_1} \times \frac{-f_2}{a}$

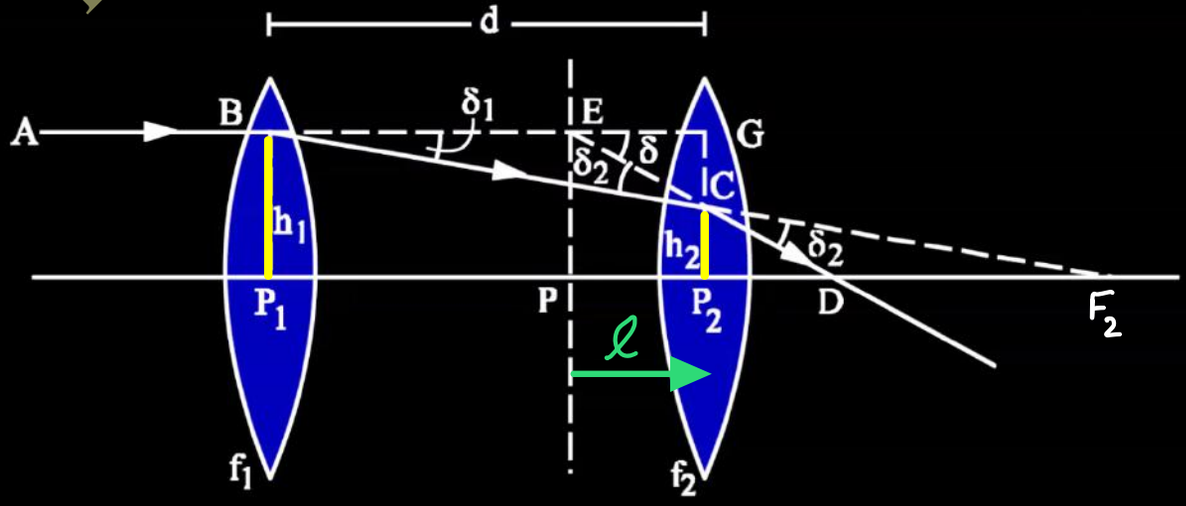
thus $\boxed{m = -f_2/f_1}$. Thus such a system produces constant magnification irrespective of object location.

→ Such an arrangement can also be made using a combination of converging and diverging lens, just that image focus of first lens must coincide with object focus of the other.

Infinity Equivalent Of Separated Doublet:

If the object is at infinity (or very large distances as compare to size and focal lengths of lens system), a combination of two thin lenses may be considered as a single thin lens. For instance, in the shown doublet, a ray incident from infinity emits along CD, thus it appears that as if, there were a single lens and refraction occurred at plane EP. Thus an effect same as that produced by this system could be produced by placing a single lens at EP plane. Now we will try to quantitative evaluate the effective focal length PD of the system as well as distance PP_2 defining the location of the equivalent lens.





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Let f_1 and f_2 be the focal lengths of the lenses and let a ray be incident at a height h_1 above the axis of the system. Then:

$$\delta = \delta_1 + \delta_2 \Rightarrow \frac{h_1}{F} = \frac{h_1}{f_1} + \frac{h_2}{f_2} \quad (1)$$ where F is effective focal length.

Now $h_2 = h_1 - d\delta_1 = h_1 - d\frac{h_1}{f_1}$ (2), now putting this in (1):

$$\frac{h_1}{F} = \frac{h_1}{f_1} + \frac{1}{f_2}\left[h_1 - \frac{d h_1}{f_1}\right] \quad (3)$$ Cancelling h_1:

$$\boxed{\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}}$$

Now $PP_2 = l = EG = \frac{(h_1 - h_2)}{\tan\delta} = \frac{d h_1}{f_1} \times \frac{1}{\delta}$ (using (2))

$= \frac{d h_1}{f_1} \times \frac{F}{h_1}$, thus $PP_2 = l = \frac{dF}{f_1}$, thus location of the lens is defined by $\boxed{l = \frac{dF}{f_1}}$

→ Note that this equivalence is valid only for object at infinity and should not be used for any other case.

Power Of A Lens: Reciprocal of focal length is called power of a lens, mathematically,

$$\boxed{P = \frac{1}{f}}$$

If focal length is substituted in metres, we get power in the unit of dioptres (D). Thus if focal length is $+0.1$ m, power is $+10$ D.

→ Converging lenses have positive power, diverging lenses have negative powers.

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Power Of A Mirror: Negative of reciprocal focal length of a mirror is called power of the mirror thus $\boxed{P_m = -\frac{1}{f_m}}$

→ Converging mirrors have positive power while diverging mirrors have negative power.

Achromatic Separated Doublet (Object At Infinity):

Recall that in a previous section we found infinity equivalent focal length of a separated doublet as:

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} \quad \text{in terms of power of lenses}$$

we can write this equation as $P = P_1 + P_2 - d P_1 P_2$ ①

when P, P_1 and P_2 are effective power, power of first lens and power of second lens respectively. Now suppose we want an achromatic combination of the two lenses (same dispersive powers), then,

$\frac{dP}{d\mu} = 0$, Now by lens maker's formula, for any lens,

power $P = (\mu-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \Rightarrow \frac{dP}{d\mu} = \left(\frac{1}{R_1} - \frac{1}{R_2}\right) = K$ (say).

Now differentiating ① wrt μ:

$$\frac{dP}{d\mu} = \frac{dP_1}{d\mu} + \frac{dP_2}{d\mu} - d\left[P_1\frac{dP_2}{d\mu} + P_2\frac{dP_1}{d\mu}\right] = 0$$

$$\Rightarrow K_1 + K_2 - d\left[P_1 K_2 + P_2 K_1\right] = 0$$

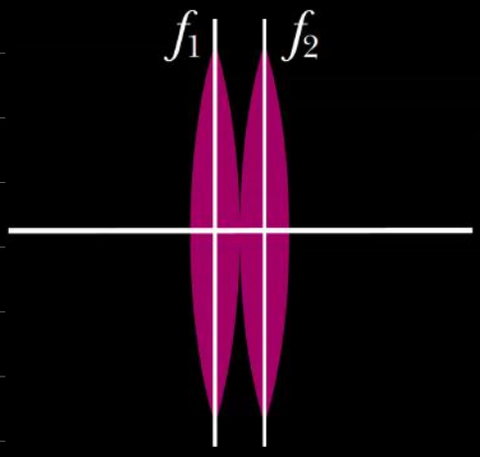
$$\Rightarrow d = \frac{K_1 + K_2}{P_1 K_2 + P_2 K_1} = \frac{K_1 + K_2}{(\mu-1)K_1 K_2 + (\mu-1)K_2 K_1} = \frac{1}{\mu-1}\left[\frac{K_1+K_2}{2K_1K_2}\right]$$

$$\Rightarrow d = \frac{1}{2(\mu-1)}\left[\frac{1}{K_1} + \frac{1}{K_2}\right] = \frac{f_1+f_2}{2}$$

Thus for achromatic doublet, $\boxed{d = \frac{f_1+f_2}{2}}$

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STACKED THIN LENSES: Whenever two thin lenses are kept in contact, their effective focal length is given as $\boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}}$ thus in terms of powers,



$\boxed{P = P_1 + P_2}$ This can be proved simply by repeated application of lens maker's formula. Let v_1 be the coordinate of intermediate image. Then:

$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$ ① now image from first lens serves as object for the second lens, now let v be the final coordinate of the image thus:

$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$ ② adding ① & ②:

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{f}$ thus $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$ Q.E.D.

→ A thin lens with same radius of curvature (signed) on both sides has infinite focal length and produces no effect on location / size of final image. This can be easily seen thus:

$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R} - \frac{1}{R}\right] = 0 \Rightarrow f = \infty$ now:

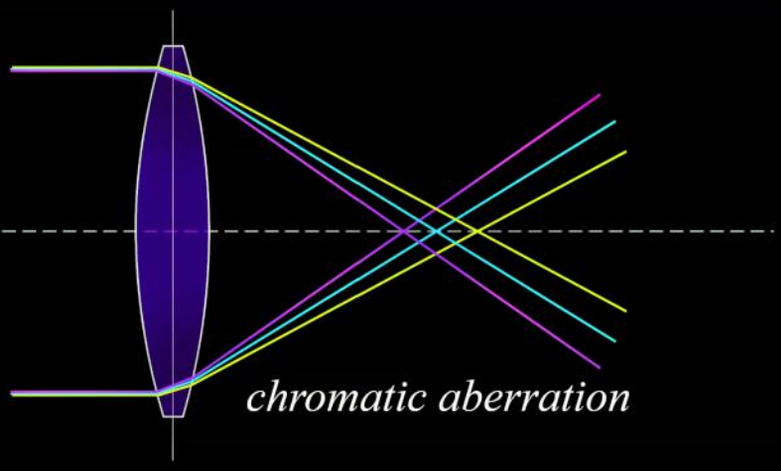
$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = 0 \Rightarrow v = u$, thus image is at same position as object.

→ This means introducing a thin film of any material on any refracting surface will have no effect on size, shape and location of final image (except for possible TIR / partial reflection effects).

→ If there are more than two lenses, the same formula can be extended thus, $P = P_1 + P_2 + \cdots$ and $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \cdots$

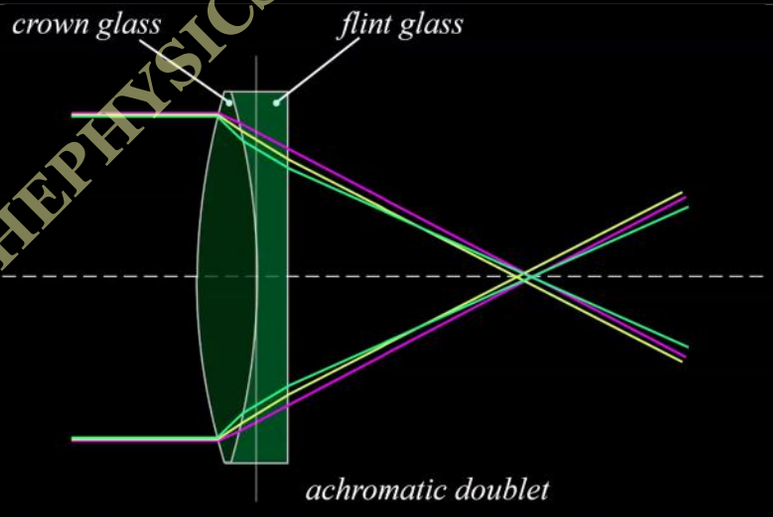
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Chromatic Aberration In Lenses:



Due to different refractive indices for different colors, the lenses have different focal lengths for different colors (smallest magnitude for violet colour because it has highest refractive index). Because of this, it is not possible to focus all the colours simultaneously in an image using a single lens. This image defect is known as chromatic aberration and can be corrected by using achromatic doublets (achromats).

Achromatic Stacked Doublet:



Using a stack of two or more lenses of different dispersive powers, it is possible to make effective focal length of a lens combination, nearly independent of wavelength (colour). Such a combination is called an achromatic combination. A typical combination used for this purpose is a stack of crown glass lens and flint glass lens. Here if f_1 is focal length of first lens (for yellow) having a dispersive power ω_1 and f_2 is the focal length of the second lens having dispersive power ω_2 then, for achromatic combination:

$$\boxed{\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0}$$, This can be shown using lens maker's formula:

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = (\mu - 1)\,K \quad \text{say.}$$

Now $\frac{1}{f_v} = (\mu_{v_1} - 1)\,K_1 + (\mu_{v_2} - 1)\,K_2$ (1) (f_v: focal length for violet, $K_1 : \frac{1}{R_1} - \frac{1}{R_2}$ for first lens etc.)

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Similarly:

$$\frac{1}{f_R} = (\mu_{R_1} - 1) K_1 + (\mu_{R_2} - 1) K_2 \quad (2) \qquad (R: \text{red}).$$

Now for achromatic combination, $\frac{1}{f_V} - \frac{1}{f_R} = 0$ thus:

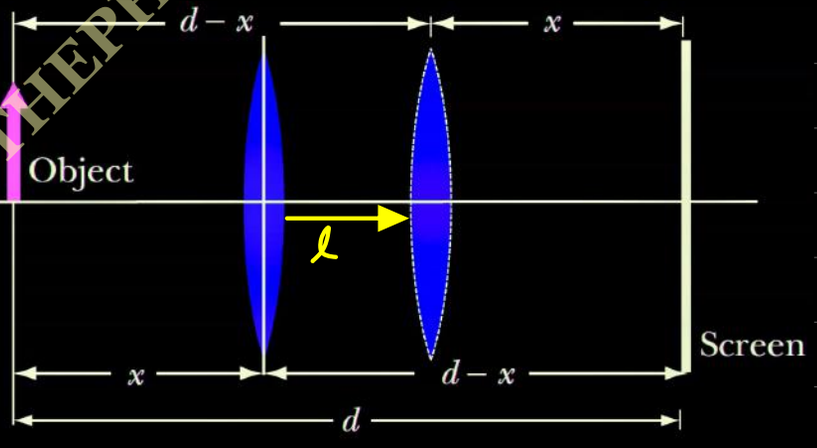
$$(\mu_{V_1} - \mu_{R_1}) K_1 + (\mu_{V_2} - \mu_{R_2}) K_2 = 0 \quad (3)$$

$$\Rightarrow \left(\frac{\mu_{V_1} - \mu_{R_1}}{\mu_1 - 1}\right)(\mu_1 - 1) K_1 + \left(\frac{\mu_{V_2} - \mu_{R_2}}{\mu_2 - 1}\right)(\mu_2 - 1) K_2 = 0 \quad (\mu: \text{Yellow}).$$

$$\Rightarrow \boxed{\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0}$$

→ If more than two lenses are stacked, by a similar logic $\Sigma \omega_i / f_i = 0$.

Displacement Method for Finding Focal Length of Convex Lens:



In this method, the object is placed at a fixed distance from a screen mounted on optical bench and a lens is moved on the bench to produce sharp image on the screen. This happens for two positions of the lens if $d > 4f$ (will be proved shortly). Knowing the distance between the object and the screen and the separation between two positions of the lens, we can find the focal length of the lens.

Consider the above figure. Let x be the lens distance for sharp image then:

$$\frac{1}{d-x} - \frac{1}{-x} = \frac{1}{f} \Rightarrow x_1 = \frac{1}{2}\left[d - \sqrt{d^2 - 4fd}\right]$$

$$x_2 = \frac{1}{2}\left[d + \sqrt{d^2 - 4fd}\right]$$

Thus to get sharp image, $d^2 - 4fd > 0 \Rightarrow \boxed{d > 4f}$

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Now $x_2 - x_1 = \sqrt{d^2 - 4fd} = \ell \quad \Rightarrow \quad \boxed{f = \frac{d^2 - \ell^2}{4d}}$

Now recognize that $x_1 = d - x_2$ and $x_2 = d - x_1$, thus we can show that magnifications of two positions are reciprocals of each other:

$$m_1 = \frac{v_1}{u_1} = \frac{d - x_1}{-x_1} = -\frac{x_2}{x_1} \quad \text{similarly } m_2 = -\frac{x_1}{x_2} \quad \text{thus } m_1 m_2 = 1$$

$\boxed{m_1 m_2 = 1}$, which means object height is geometric mean of two image heights i.e $\boxed{h_o = \sqrt{h_{i_1} h_{i_2}}}$, this can be readily seen:

$$m_1 m_2 = 1 \Rightarrow \frac{h_{i_1}}{h_o} \times \frac{h_{i_2}}{h_o} = 1 \Rightarrow h_o = \sqrt{h_{i_1} h_{i_2}} \quad \text{QED.}$$

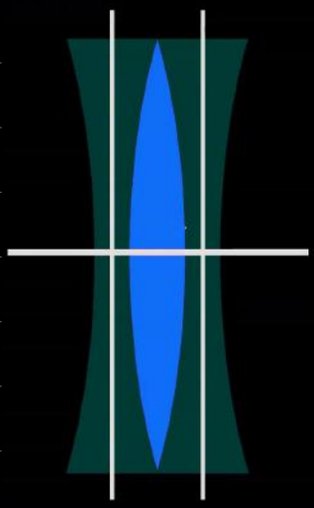
EXAMPLE: In an experiment to find focal length of a convex lens, the object and the screen are fixed. The separation between two positions of lens for which sharp image is obtained is 20 cm. Magnification for one of the positions of lens is -2. Find the focal length of the lens.

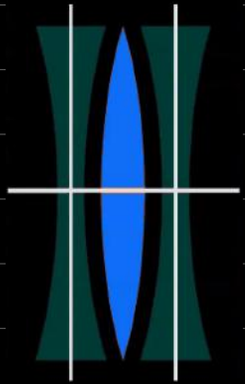
Solution: Using the given data: $\left(\frac{d-x}{x}\right) = 2$ and $(d-x) - x = 20$

$\Rightarrow \quad d - 3x = 0 \quad$ and $\quad d - 2x = 20 \quad \Rightarrow \quad x = 20 \quad$ and $\quad d = 60$ cm

Now $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{40} - \frac{1}{-20} = \frac{1}{f} \quad \Rightarrow f = +\frac{40}{3}$ cm Ans.

EXAMPLE: Two equiconcave lenses of focal length 20 cm each having a refractive index 1.5 are kept together and space between them is filled with water as shown in the figure. Find the equivalent focal length of the arrangement.





Solution: As discussed earlier, by introduction of a thin air film, the system effectively remains the same. However this enables us to visualise the system as combination of three thin lenses with symmetric media (air) on both sides. Now let radius of curvature of curved surfaces

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have a magnitude R, then:

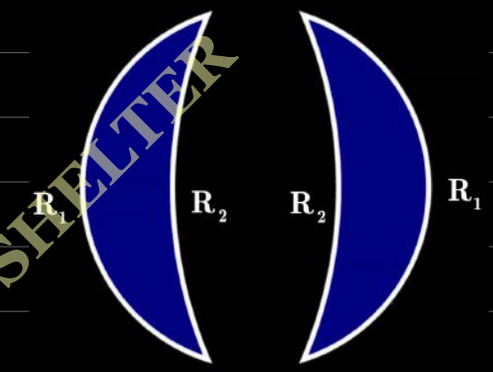
$$\frac{-1}{20} = (1.5-1)\left[\frac{1}{-R} - \frac{1}{R}\right] \Rightarrow R = 20\,cm,$$ Now focal length of

water lens: $$\frac{1}{f_w} = \left(\frac{4}{3}-1\right)\left[\frac{1}{20} - \frac{1}{-20}\right) = \frac{1}{30} \Rightarrow f_w = +30\,cm$$

Now effective focal length of the system: $\frac{1}{f} = \Sigma \frac{1}{f_i}$ thus:

$$\frac{1}{f} = \frac{1}{-20} + \frac{1}{30} + \frac{1}{-20} = \frac{-3+2-3}{60} = \frac{-1}{15} \Rightarrow f = -15\,cm$$ Ans.

→ Flipping of a lens does not make any difference to its focal length for instance here, for the left lens: $\frac{1}{f} = (\mu-1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$ while for the right lens:

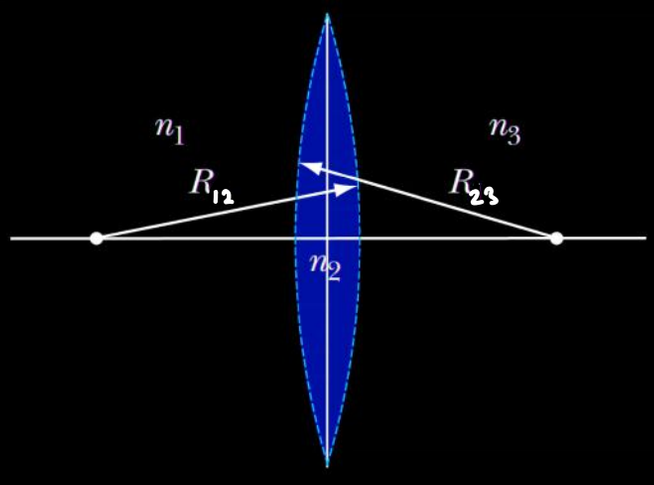


$$\frac{1}{f} = (\mu-1)\left[\frac{-1}{R_2} - \frac{1}{-R_1}\right]$$

$$\Rightarrow \frac{1}{f} = (\mu-1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$ which is the same expression as above.

Focal Length Of Lens With Different Media On Both Sides:

Consider the lens shown in adjacent figure. The lens formula for such a lens can be obtained by repeatedly applying spherical surface formula. Let u be the coordinate of object and v be the final coordinate of the image and v_1 be the coordinate of intermediate image then: For first interface:



$$\frac{n_2}{v_1} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_{12}} \quad (1)$$

Similarly for second interface:

$$\frac{n_3}{v} - \frac{n_2}{v_1} = \frac{n_3 - n_2}{R_{23}} \quad (2)$$

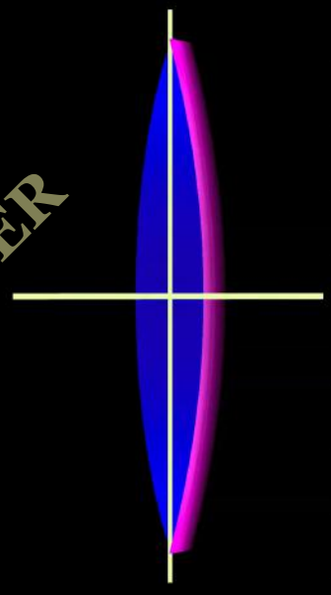
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Adding ① & ②

$$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_3 - n_2}{R_{23}} + \frac{n_2 - n_1}{R_{12}}$$

Same pattern can be continued if there are more than three media involved so long as the lens combination is still reasonably thin.

Silvered Lens / Lens Mirror Stack:



Consider the silvered lens shown in adjacent figure. If f_l is the focal length of the lens alone and f_m is the focal length of mirror alone then it can be shown that such a combination behaves effectively like a mirror with effective focal length being given as:

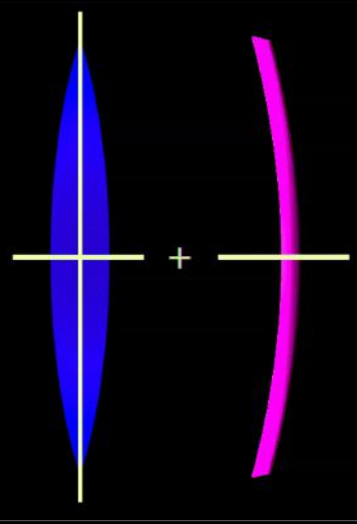
$$\frac{1}{f_{M_{eff}}} = \frac{1}{f_m} - \frac{2}{f_l}$$

or in terms of Power:

$$P_{M_{eff}} = 2P_l + P_m$$

→ If multiple lenses are stacked in front of the silvered lens then f_l in above formula needs to be replaced by effective focal length of all the lenses combined together.

Proof Of Silvered Lens Formula:



Recall that introduction of thin film of any material does not make any difference to the location and size of image thus we can imagine a very thin layer of air between silvered surface and glass to model the silvered lens as a combination of lens and a mirror. Now the optical events to be considered are: (I) Refraction from lens (II) Reflection from mirror (III) Refraction from lens (Reversed incident light direction). Let coordinate of intermediate images in original coordinate system be v_1 and v_2 then, for first event:

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_l} \quad ①$$

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For second event,

$$\frac{1}{v_2} + \frac{1}{v_1} = \frac{1}{f_m} \quad (2)$$

Now let the intermediate image coordinate in reversed coordinate be $\tilde{v}_2$ and final image coordinate in this system be $\tilde{v}$ then, for third event:

$$\frac{1}{\tilde{v}} - \frac{1}{\tilde{v}_2} = \frac{1}{f_l} \quad (3) \text{ but } \tilde{v} = -v \text{ and } \tilde{v}_2 = -v_2$$

(∵ coordinate system toggles the signs along horizontal direction. Here v is the final image coordinate in original coordinate system).

Thus (3) can be written as:

$$\frac{-1}{v} + \frac{1}{v_2} = \frac{1}{f_l} \quad (4)$$

Now: (2) − (1) − (4) gives:

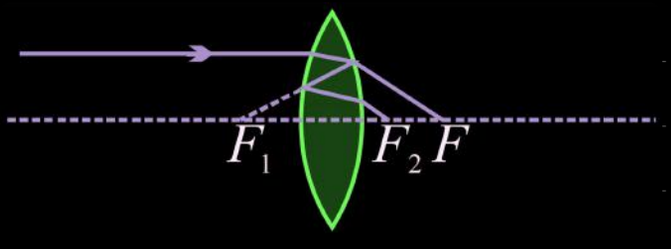
$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f_m} - \frac{2}{f_l} = \frac{1}{f_{m_{eff}}}$$

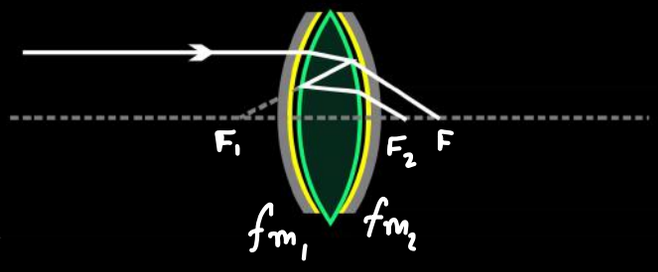
thus $\frac{1}{f_{m_{eff}}} = \frac{1}{f_m} - \frac{2}{f_l}$ QED also multiplying by -1:

$$P_{m_{eff}} = P_m + 2P_l \quad \text{QED.}$$

Image Formation Due To Multiple Internal Reflection In Lens:

Consider the ray diagram in adjacent figure Here Ray is focused at F_2 after two internal reflections. Such a lens can be modeled as a combination of a lens and two partially silvered mirrors. For F_1 we imagine two processings from the lens and one processing from mirror fm_2 thus, $P_{m_{eff}} = P_{m_2} + 2P_l$





While for F_2, we visualize 3 processings from lens and one processing each from front and back mirrors, since final image is formed after

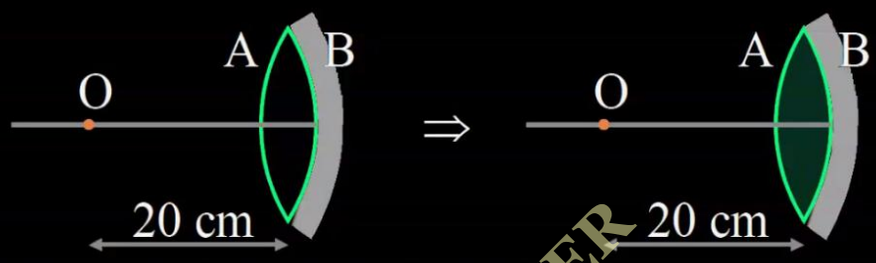
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crossing the lens, the effective combination behaves as a lens with

$$P_{L_{eff}} = 3P_L + P_{m_1} + P_{m_2}$$

EXAMPLE:

Two identical thin watch glasses are joined together with air inside as shown in figure. The surface B is silvered. A sharp image of object placed on optic axis at 20 cm from the lens is formed at distance 10 cm from lens. Find the position of the sharp image when the space between the glasses is filled with water ($\mu_w = 4/3$) (Assume real image in first case).



Solution: With air in the gap, the system is effectively equivalent to just the silvered mirror. Let radius of curvature of mirror be R then:

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R} \Rightarrow \frac{1}{-10} + \frac{1}{-20} = \frac{2}{R} = \frac{1}{f_m} \Rightarrow f_m = -\frac{20}{3}\text{ cm},\ R = -\frac{40}{3}\text{ cm}.$$

Now for the water lens:

$$\frac{1}{f_L} = \left(\frac{4}{3} - 1\right)\left[\frac{1}{40/3} + \frac{1}{40/3}\right] = \frac{1}{20} \Rightarrow f_L = 20\text{ cm}$$

Now $\frac{1}{f_{Meff}} = \frac{1}{f_m} - \frac{2}{f_L} = \frac{-3}{20} - \frac{2}{20} = \frac{-5}{20} = -\frac{1}{4} \Rightarrow f_{Meff} = -4\text{ cm}.$

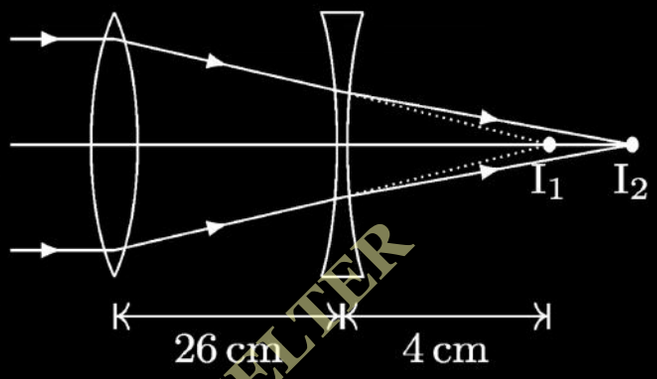
Now using mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-20} = \frac{1}{-4}$

$\Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{4} = \frac{-1}{5} \Rightarrow v = -5 \Rightarrow$ final image is 5 cm from the lens on the left side.

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Example: JEE : 2003:

The size of the image of an object, which is at infinity, as formed by a convex lens of focal length 30 cm is 2 cm. If a concave lens of focal length 20 cm is placed between the convex lens and the image at a distance of 26 cm from the convex lens, calculate the new size of the image.



Solution: Image formed by first lens serves as object for the second lens thus for concave lens: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{v} - \frac{1}{4} = \frac{1}{-20} \Rightarrow \frac{1}{v} = \frac{1}{5}$

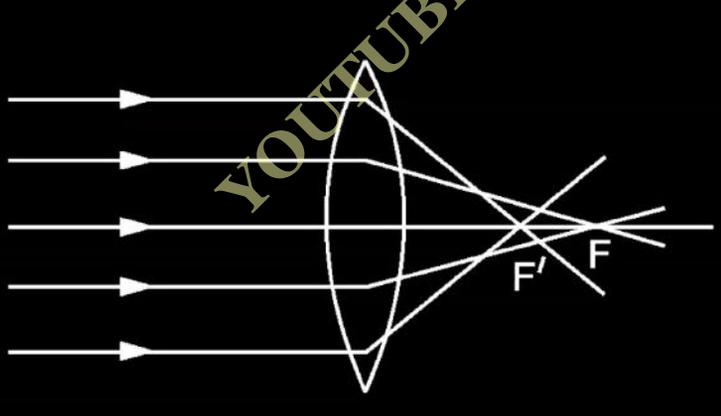
$\Rightarrow v = +5$ cm. Now magnification:

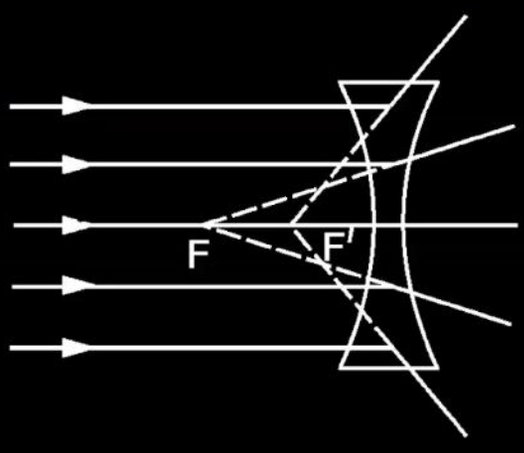
$m = v/u = 5/4 = 1.25$ thus new size of image:

$1.25 \times 2 = 2.5$ cm Ans.

→ Even though sun is at infinity its image is not a point because it subtends a finite angle at the optical centre. Here the angle subtended by the object and image at optical centre are equal.

Spherical Aberration From Spherical Lenses:





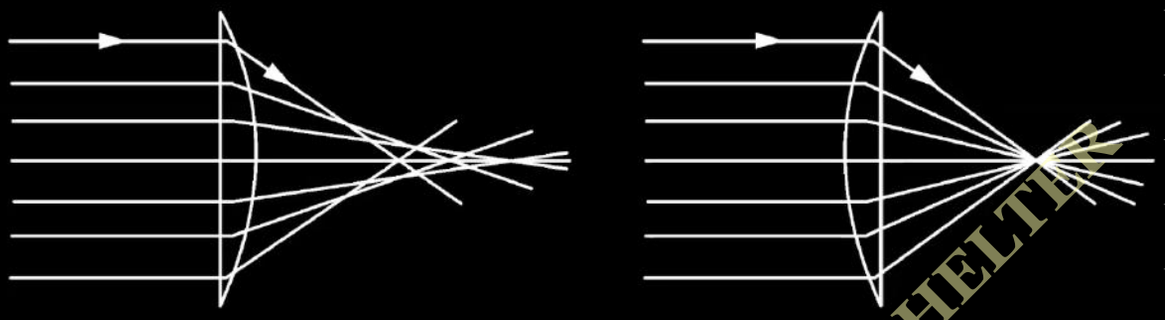
Just as with mirrors, in spherical lenses also marginal rays do not converge at the same point on principal axis as paraxial rays. Thus with a lens of wide aperture a blur image is formed. To avoid or reduce the aberration effects, pupils and stops are

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used. (Pupils block marginal incident rays while stops block the refracted rays).

→ The reason we are able to read better with bright light is because our pupil contracts to a smaller aperture and hence reduces the spherical aberration effects.

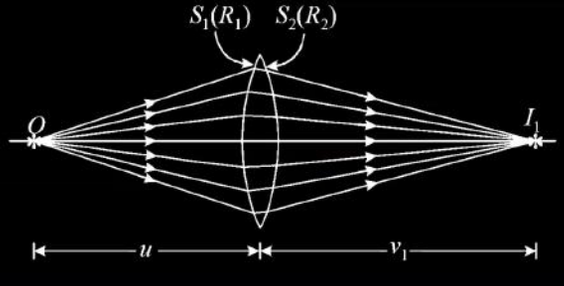
Avoiding Spherical Aberration With Plano Convex Lens:

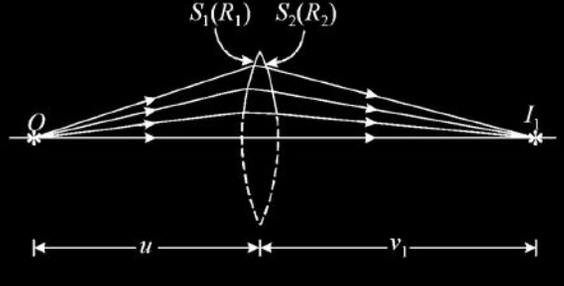


Even though we showed that flipping a lens has no effect on its focal length, however if spherical aberration is taken in to account, it can be shown that there is more spherical aberration if rays are incident on flat surface first. (In this case entire deviation of rays has to happen on the second surface). This has something to do with the fact that $\sin\theta \approx \theta$ is a better approximation for smaller angles. When light is first made to fall on convex surface, there are two smaller deviations on both refracting surfaces rather than just one relatively larger deviation on convex surface.

Cutting Or Blackening Of Portion Of A Lens:

Whenever a portion of lens is cut off by a plane parallel to principal axis, it usually produces no effect on size and location of the final image so long as the remaining portion of the lens is within the field of view of the object. However, the brightness of the image usually decreases because now the image is formed by lesser number of rays (or smaller light collecting area of the lens). Similar phenomenon is





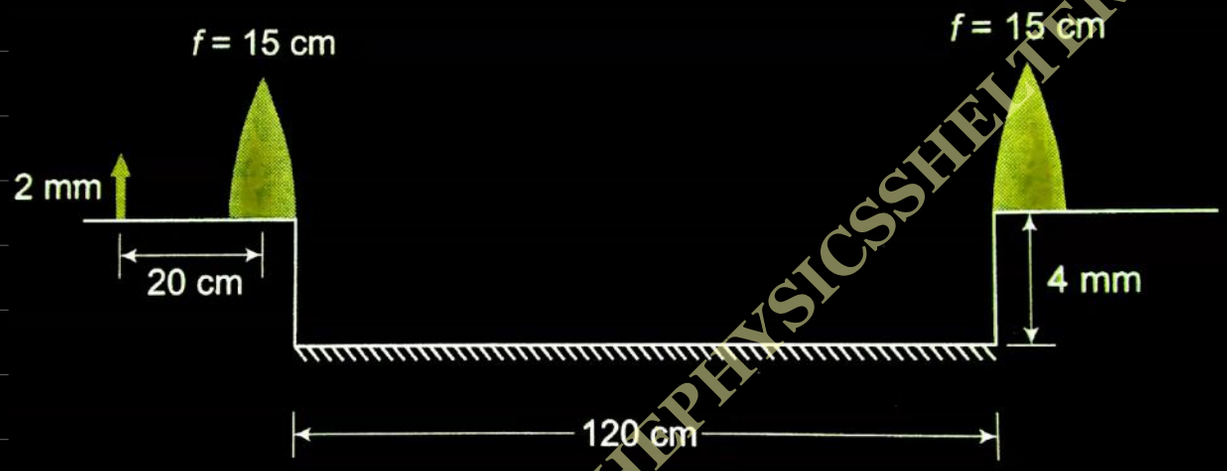
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observed due to blackening the portions of a lens.

→ To avoid making any silly mistake, very carefully visualize the principal axis by completing the lens (either mentally or actually on the papers).

EXAMPLE:

A convex lens of focal length 15 cm is split into two halves and the two halves are placed at a separation of 120 cm. Between, the two halves of convex lens a plane mirror is placed horizontal and at a distance of 4mm below the principal axis of the lens halves. An object of length 2mm is placed at a distance of 20cm from one half lens as shown in figure. Find the position and size of the final image. (Consider intermediate image formed by plane mirror).

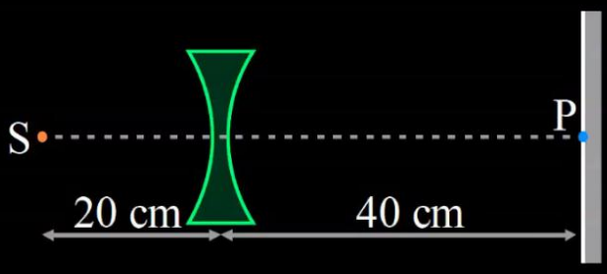


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Solution: For first lens, $xy = -f^2 \Rightarrow -5xy = -225 \Rightarrow y = 45$ cm and $m = -y/f = -45/15 = -3$ thus intermediate image from the lens is formed 60 cm and has a depth of 6mm. (i.e 2mm below the plane mirror. Thus intermediate image formed from plane mirror has its tip 2mm above the mirror and foot 4mm below the mirror. It has a height of 6mm (erect). Now this serves as object for the second lens. Now for second lens, $u = -60$ cm $\Rightarrow x = -45$ $\Rightarrow$ once again, $y = +5$ thus final image is formed 20 cm to the right of second lens and has a magnification $-1/3$. Now the intermediate image from plane mirror has its tip 2mm below the principal axis and foot 8mm below the principal axis, thus for final image, the tip is 2/3 mm above the principal axis and foot is 8/3 mm above the principal axis. Thus size of image is 2mm (inverted lying between 2/3 mm and 8/3 mm above principal axis) and located at 20 cm to the right of second lens.

Example:

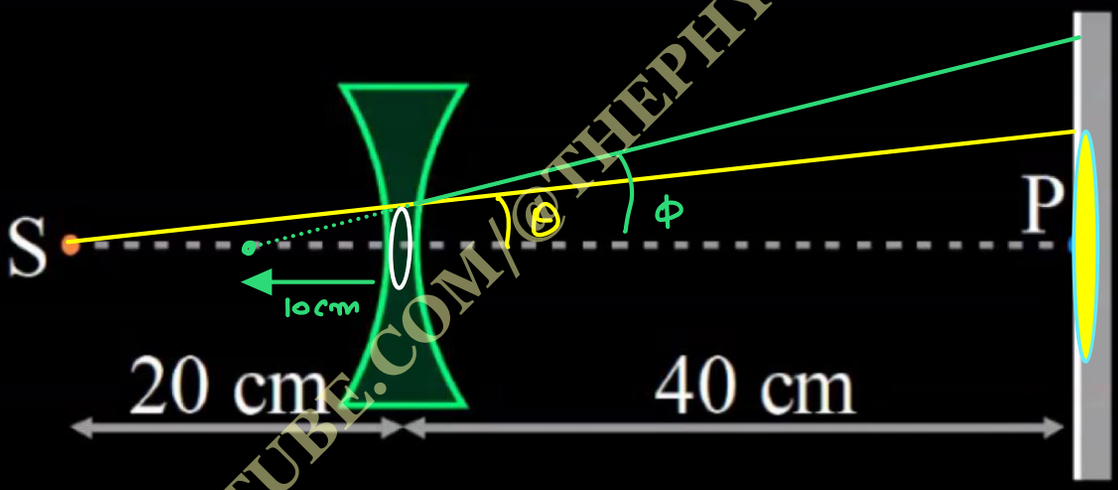
A point source of light is placed 60cm away from screen. Intensity detected at point P is I. Now a diverging lens of focal length 20 cm is placed 20 cm away from S between S and P. The lens transmits 75% of light incident on it. Find the new value of intensity at P.



(A) 0.21 I (B) 0.27 I (C) 0.33 I (D) 0.37 I

Solution: Due to diverging lens, the light that was falling on a certain patch near P will fall on a larger patch, thus decreasing the intensity. For the diverging lens, $u = -20$, $f = -20$ thus:

$$\frac{1}{v} - \frac{1}{-20} = \frac{1}{-20} \Rightarrow v = -10 \text{ cm}.$$



Now consider the ray diagram above. Earlier whatever light was spread in a circular spot of radius $60\tan\theta$, is now spread in a circle of radius $80\tan\phi$, thus new intensity:

$$I' = \eta I \left(\frac{60\tan\theta}{80\tan\phi}\right)^2,$$ Now let r be the radius of circle on the lens through which all this light is passing then:

$r = 10\tan\phi = 20\tan\theta \Rightarrow \tan\theta/\tan\phi = 1/2$

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Thus $I' = I \times 0.75 \times \left(\frac{6}{5} \times \frac{1}{2}\right)^2 = 0.27\,I$ Ans.

Alternate Solution:

Power recieved by any patch = (Power Emitted / unit solid angle) × Solid Angle of the patch.

Now consider a very small circular region near centre of lens. At this circle, power emitted per unit solid angle gets scaled down by a ratio of new solid angle to old solid angle i.e $(S/10^2) : (S/20^2)$ where S is the area of circular region at the lens.

Thus power emitted / unit solid angle becomes $\frac{1}{4}$ times.

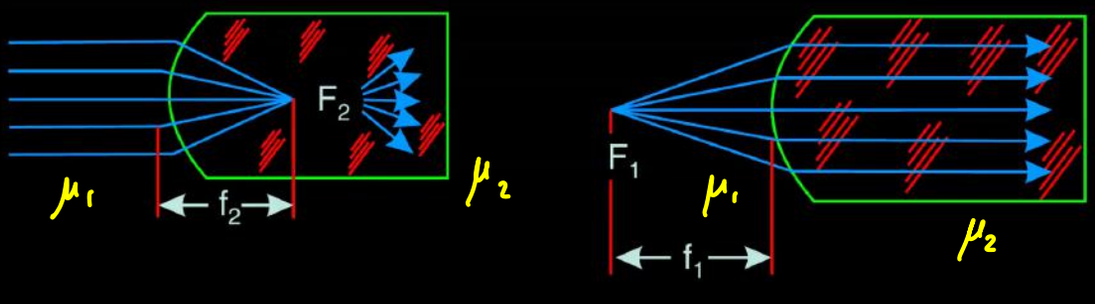
Now let S be a small patch area near P, then solid angle at the centre of emission get scaled up by a factor of $\frac{S}{50^2} : \frac{S}{60^2}$

i.e $\left(\frac{6}{5}\right)^2$ times. Thus intensity gets scaled by a factor of

$$\frac{1}{4} \times \left(\frac{6}{5}\right)^2 \times 0.75 = 0.27\,I \text{ Ans.}$$

REDUCED DISTANCE: Any distance divided by the refractive index of the medium in which that distance is measured is called reduced distance. Thus we can talk about reduced focal length if processed rays are emitted in to a medium of refractive index μ. If f is the focal length in a medium of refractive index μ then, f/μ is the reduced focal length.

POWER OF REFRACTING SPHERICAL SURFACE:



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Recall the spherical surface formula: $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

Now when $u = -\infty$, $v = f_2$, thus, $\frac{\mu_2}{f_2} = \left(\frac{\mu_2 - \mu_1}{R}\right)$

$\Rightarrow \frac{\mu_2}{f_2} = \left(\frac{\mu_2 - \mu_1}{R}\right)$ now since f_2 is defined by refracted ray passing in to medium 2 thus reduced focal length is $\frac{f_2}{\mu_2}$ accordingly power of the spherical surface is $\frac{\mu_2}{f_2} = (\mu_2 - \mu_1)/R$

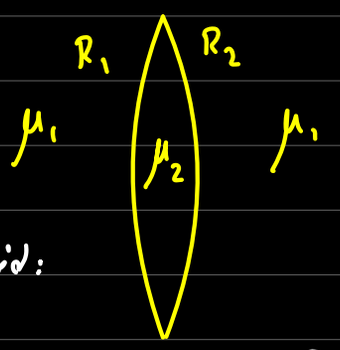
Thus $\boxed{P = \frac{\mu_2}{f_2} = \frac{(\mu_2 - \mu_1)}{R}}$

Power Of An Immersed Mirror: If a mirror reflects in to a medium of refractive index μ, its Power is $-\mu/f$ where f is its focal length.

Power Of An Immersed Lens:

Recall that for lens with asymmetric medic on both sides,

$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_3 - \mu_2}{R_{23}} + \frac{\mu_2 - \mu_1}{R_{21}} \text{ etc.}$$



Now applying this to the lens immersed in liquid:

$$\frac{\mu_1}{v} - \frac{\mu_1}{u} = \frac{\mu_1 - \mu_2}{R_2} + \frac{\mu_2 - \mu_1}{R_1} = (\mu_2 - \mu_1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

Now $\frac{1}{v} - \frac{1}{u}$ is $\frac{1}{f_l}$ where f_l represents focal length in the liquid and the RHS is nothing but summation of Powers of two surfaces. Thus we can now write:

$P_l = P_1 + P_2$ where P_1 and P_2 are powers of spherical surfaces.

$$\boxed{P_l = P_1 + P_2}$$

$$\boxed{\mu_1\left[\frac{1}{v} - \frac{1}{u}\right] = (\mu_2 - \mu_1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]}$$

$$\frac{\mu_1}{f_l} = (\mu_2 - \mu_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = P_1 + P_2$$

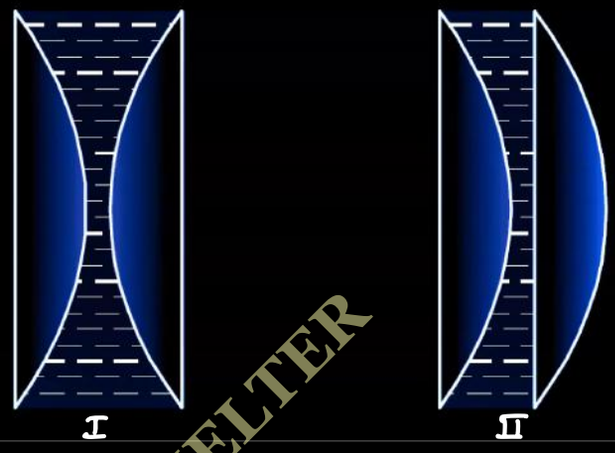
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→ The above formula can be conveniently used for refraction through several spherical surfaces when medium before first refraction and after last refraction are same.

EXAMPLE: In the shown systems, the magnitude of radius of curvature of curved surfaces is 10cm and the materials are water and glass.

(a) Find effective focal length of the two systems in air

(b) Find effective focal lengths of two systems if immerse in a liquid of $\mu = 2$.



Solution: (a)$_I$: $\frac{1}{f} = \frac{3/2 - 1}{\infty} + \frac{4/3 - 3/2}{-10} + \frac{3/2 - 4/3}{10} + \frac{1 - 3/2}{\infty}$

$\Rightarrow \frac{1}{f} = \frac{1}{30} \Rightarrow f = +30\ cm$ Ans.

(a)$_{II}$ $\frac{1}{f} = \frac{3/2 - 1}{\infty} + \frac{4/3 - 3/2}{-10} + \frac{3/2 - 4/3}{\infty} + \frac{1 - 3/2}{-10}$

$\Rightarrow \frac{1}{f} = \frac{1}{60} + \frac{1}{20} = \frac{1}{15} \Rightarrow f = +15\ cm.$

(b)$_I$ $\frac{2}{f} = \frac{3/2 - 2}{\infty} + \frac{4/3 - 3/2}{-10} + \frac{3/2 - 4/3}{10} + \frac{2 - 3/2}{\infty}$

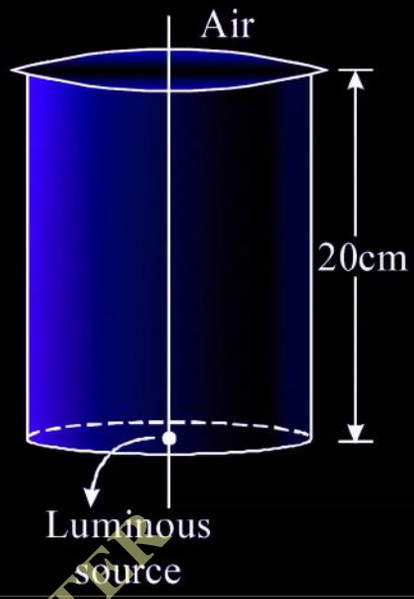
$\Rightarrow \frac{2}{f} = \frac{1}{60} + \frac{1}{60} = \frac{1}{30} \Rightarrow f = +60\ cm.$

(b$_{II}$) $\frac{2}{f} = \frac{3/2 - 2}{\infty} + \frac{4/3 - 3/2}{-10} + \frac{3/2 - 4/3}{\infty} + \frac{2 - 3/2}{-10} = \frac{1}{60} - \frac{1}{20} = \frac{-1}{30}$

$\Rightarrow f = -60\ cm$ Ans. (Acts like diverging lens).

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Example: An equiconvex glass lens is placed on top of a vessel of height 20 cm as shown in the figure. A luminous source is placed at the bottom of the vessel. When vessel is empty, image of source is formed 20cm above the lens. When vessel is completely filled with a liquid of refractive index μ, the image of the source is formed 30 cm above the lens. Find the refractive index of the liquid.



Solution: From first information, $2f = 20\text{ cm} \Rightarrow f = 10\text{ cm}$.

Thus $\frac{1}{10} = \left(\frac{3}{2}-1\right)\left(\frac{1}{R}+\frac{1}{R}\right) \Rightarrow R = \pm 10\text{ cm}$ for both surfaces.

Now applying formula for lens with asymmetric media on both sides,

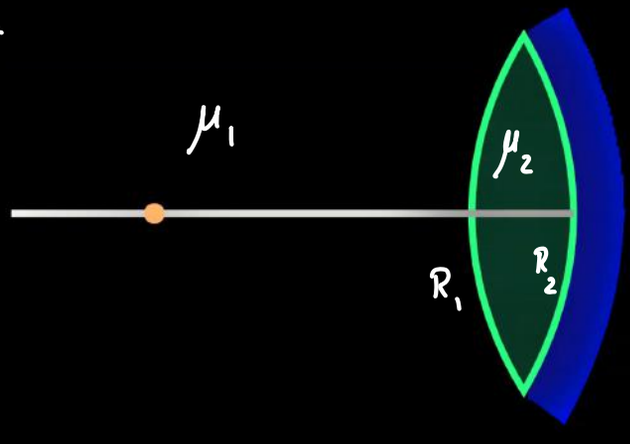
$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_3-\mu_2}{R_{23}} + \frac{\mu_2-\mu_1}{R_{21}}$ thus: $\frac{1}{30} - \frac{\mu}{-20} = \frac{1-3/2}{-10} + \frac{3/2-\mu}{10}$

$\Rightarrow \frac{\mu}{20} + \frac{\mu}{10} = \frac{1}{20} + \frac{3}{20} - \frac{1}{30} \Rightarrow \frac{3\mu}{20} = \frac{3+9-2}{60} = \frac{1}{6}$

$\Rightarrow \mu = 10/9$ Ans.

Silvered Lens Immersed In A Liquid:

Consider a silvered lens being immersed in a liquid. The events that happen are refraction at the front surface followed by reflection from mirror and then another refraction (reversed coordinate system). Now considering the events one by one:



$$\frac{\mu_2}{v_1} - \frac{\mu_1}{u} = \frac{\mu_2-\mu_1}{R_1} \quad (1)$$

(v_1 is first intermediate).

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Now for mirrors:

$$\frac{1}{v_2} + \frac{1}{v_1} = \frac{1}{f_m} \quad (2)$$

Now for refraction at curved surface (reversed coordinates):

$$\frac{\mu_1}{v} - \frac{\mu_2}{v_2} = \frac{\mu_1 - \mu_2}{R_1}$$ (Barred coordinate in the reversed system).

$$\Rightarrow \quad -\frac{\mu_1}{v} + \frac{\mu_2}{v_2} = \frac{\mu_2 - \mu_1}{R_1} \quad (3)$$

Now (1) $\times -1$ + (3) $\times (-1)$ + (2) $\times \mu_2$ gives:

$$\boxed{\frac{\mu_1}{v} + \frac{\mu_1}{u} = \frac{\mu_2}{f_m} - \frac{2(\mu_2 - \mu_1)}{R_1}}$$

Now $\frac{1}{v} + \frac{1}{u} = \frac{1}{f_e}$, thus system behaves like a single mirror immersed in a liquid of refractive index μ_1 and in terms of Powers we can write the above equation as:

$$\boxed{P_e = P_m + 2P_R}$$ where P_R is power of refracting surface.

→ This also gives us an alternative way to deal with silvered lenses in air. In case of multiple refractions, all the powers of refracting surfaces need to be added up.

→ Note that for P_m, reflection from mirror happens in μ_2 $\therefore P_m = -\frac{\mu_2}{f_m}$

while $P_{m_{eff}} = -\mu_1 / f_{M_{eff}}$.

EXAMPLE: A biconvex thin lens is prepared from glass of refractive index 3/2. The bounding surfaces have radii 25cm each. The lens is silvered as shown. Where should an object be placed before this lens so that image coincides with object.



Solution: Method I: (Previous section): The object must be placed at $2f_{M_{eff}}$. Now:

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$$\frac{\mu_1}{f_e} = \frac{\mu_2}{f_m} - \frac{2(\mu_2-\mu_1)}{R_1} \quad \text{thus:}$$

$$\frac{1}{f_e} = \frac{3/2}{-12.5} - 2\left[\frac{3/2-1}{25}\right] \Rightarrow \frac{1}{f_e} = \frac{-3}{25} - \frac{1}{25} = \frac{-4}{25}$$

$\Rightarrow f_e = -25/4 \Rightarrow$ location of object $= 2f_e = -25/2 = -12.5$ cm Ans.

Method II: Breaking in to lens + Mirror: $\frac{1}{f_e} = \left(\frac{3}{2}-1\right)\left[\frac{1}{25}+\frac{1}{25}\right]$

$\Rightarrow f_e = 25$ cm, Now $f_m = -12.5$ cm thus $\frac{1}{f_e} = \frac{-1}{12.5} - \frac{2}{25} = \frac{-4}{25}$

thus $f_e = -25/4$ cm (this is the same result that we got before).

Think: For the silvered lens, suppose it is immerse in liquid of refractive index 2. Now where should you place an object for auto collimation.

Solution: $\mu_1\left[\frac{1}{u}+\frac{1}{v}\right] = \frac{\mu_2}{f_m} - \frac{2(\mu_2-\mu_1)}{R}$, thus:

$$2\left[\frac{1}{u}+\frac{1}{u}\right] = \frac{3/2}{-12.5} - \frac{2[2-1]}{+25} \Rightarrow \frac{4}{u} = \frac{-3}{25} - \frac{2}{25} = \frac{-1}{5}$$

$\Rightarrow u = -20$ cm. Thus object should be placed 20 cm in front of the silvered lens.

EXAMPLE: PATHFINDER: MCQ 6:

A lens held above a coin placed on a table forms an image of the coin. After the lens is moved vertically a distance equal to its focal length, it forms another image of the coin equal in size to the previous image. If diameter of the coin is 2.0 cm, what is the diameter of the image?

(a) 3.0 cm. (b) 4.0 cm
(c) 5.0 cm (d) 6.0 cm

Solution: Recall that $xy = -f^2$ and $m = f/x$, now since magnifications in two position have equal magnitudes initially coin would have been some distance above the focus and later it must be

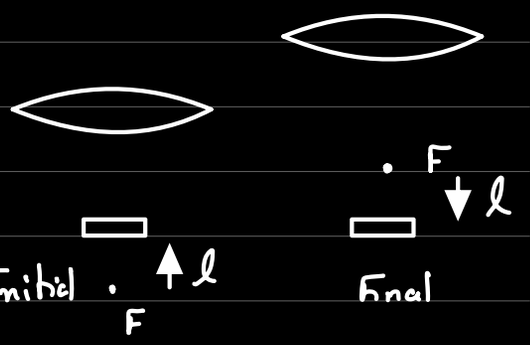
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https://www.youtube.com/@ambarishphysics https://www.youtube.com/@thephysicsshelter

same distance below the focus (say a distance l). Thus, Thus lens must have moved through $2l$.

Thus $2l = f \Rightarrow l = f/2$, Now magnification $= f/n$ thus $|m| = f/(f/2) = 2$

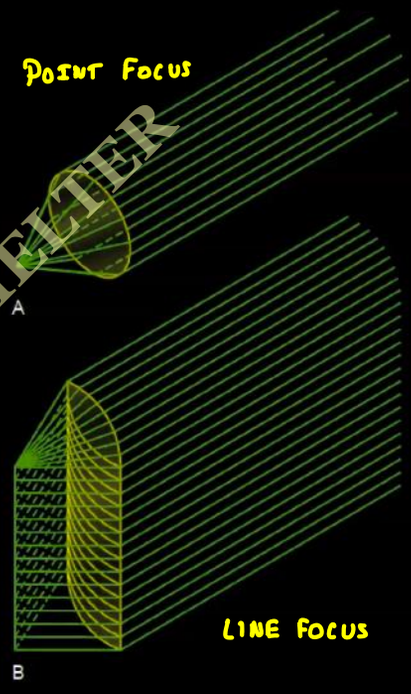
Thus image diameter must be 2×2 i.e 4cm.

Option (b). Ans.



Cylindrical Lenses:

Cylindrical lenses have infinite radius of curvature in one direction while a finite radius of curvature in the perpendicular direction. They do not cause convergence along the direction where radius of curvature in infinite. Unlike spherical lenses, they converge a parallel beam of light in to a line. Cylindrical lenses are mostly used in spectacles for people having astigmatism (eye lens having cylindrical defect where, the lens have different radii of curvature in different directions).

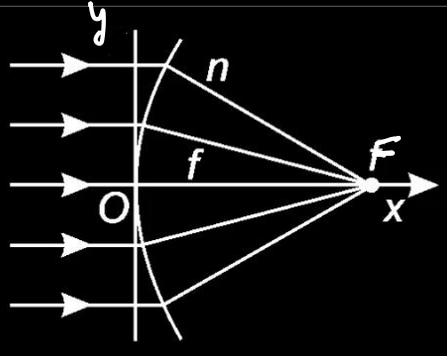


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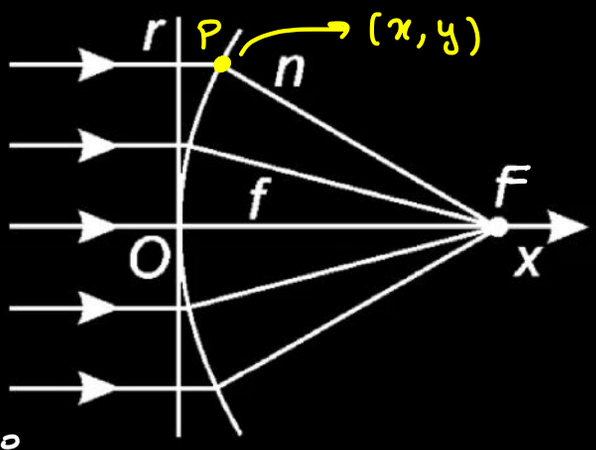
Fermat's Principle: The path taken by a ray between two given points is the path that can be travelled in least possible time. If rays are taking multiple paths between two points in that case all those paths are the paths of minimum (degenerate) time or all the paths will require same travel time.

Example: Irodov: 5.30:

A parallel beam of light falls from vacuum on a surface enclosing a medium with refractive index n (See Fig.). Find the shape of that surface, $x(y)$, if the beam is brought into focus at the point F at a distance f from the crest O. What is the maximum radius of a beam that can still be focussed?



<u>Solution:</u> By fermat's principle time taken by the rays to reach point F must be same for all the rays. Now let (x, y) be a point on the refracting surface, then $\frac{OF}{c/\mu} = \frac{x}{c} + \frac{PF}{c/\mu}$

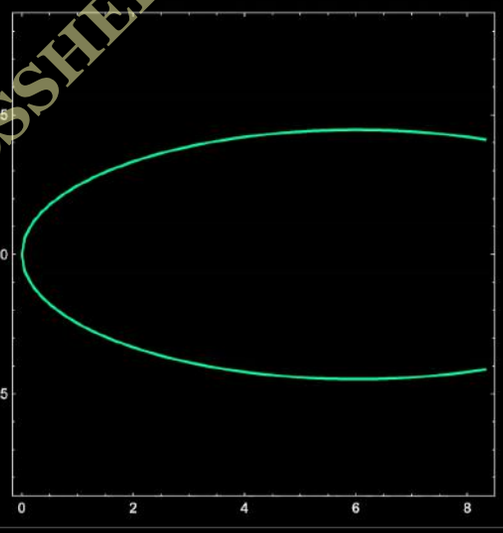


$\Rightarrow \mu(OF) = x + \mu PF$ or

$\mu f = x + \mu\sqrt{(f-x)^2+y^2}$ ①, $x>0, f>0$

This is the required implicit equation of the refracting surface.

A typical plot of the curve is as shown in the adjacent figure (for $\mu = 1.5$, $f = 10$cm). Clearly the slope of this curve becomes zero at some point which would imply 90° angle of incidence, beyond which, no parallel ray can be incident directly. Thus maximum radius of the beam that can be focused can correspond to its incidence at a point when slope of the curve becomes zero. To locate this point, we will differentiate equation ① and put $dy/dx = 0$ thus:



$$0 = 1 + \mu\left[\frac{1}{2\sqrt{(f-x)^2+y^2}} \cdot \left[2(f-x)(-1) + 2\,dy/dx\right]\right]$$

$dy/dx \to 0$

Thus: $\mu(f-x) = \sqrt{(f-x)^2+y^2}$ ②

Now we need to solve ① & ② for y, solving we get:

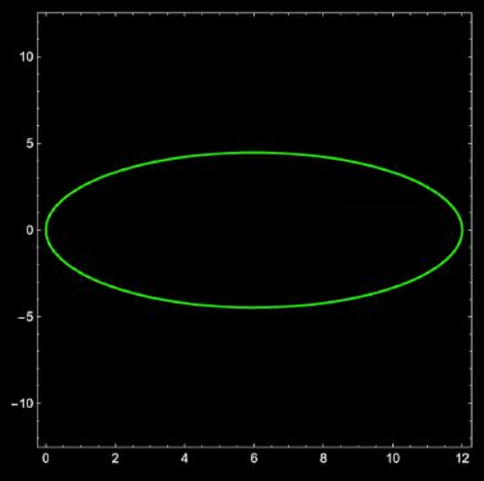
$x = \frac{\mu f}{\mu+1}$ and $y = \sqrt{\frac{\mu-1}{\mu+1}}\, f = R_{max}$ Ans.

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→ Note that equation ① represents an ellipse:
If we solve ① for x, we get:

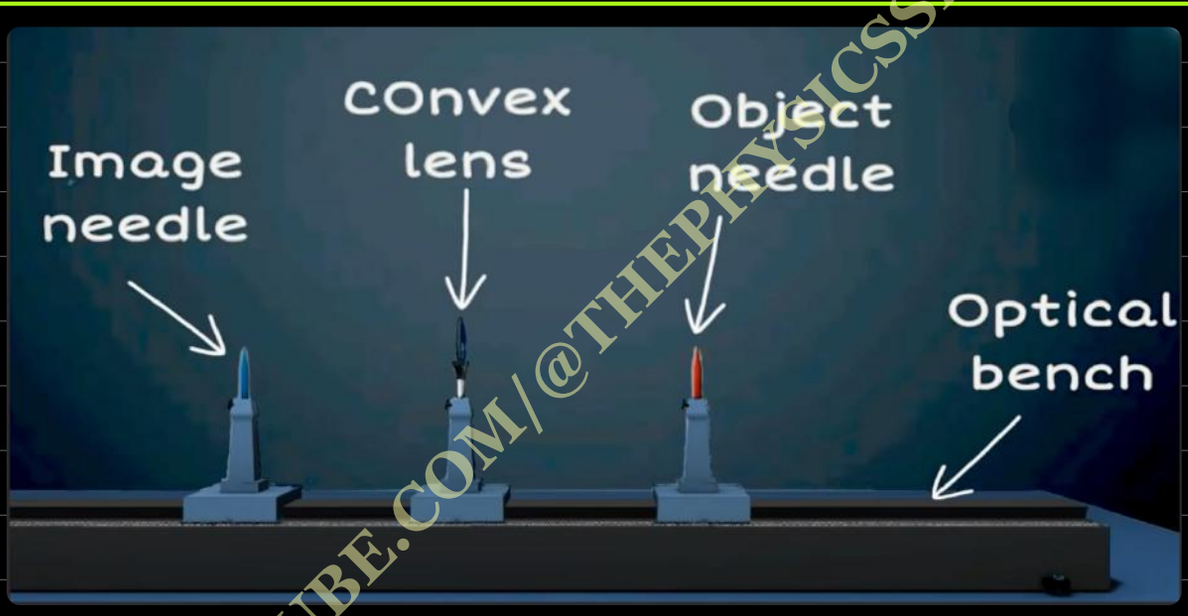
$$x = \frac{\mu f}{\mu+1}\left[1 \pm \sqrt{1-\frac{(\mu+1)}{(\mu-1)}\frac{y^2}{f^2}}\right]$$

we are interested only in its front portion so our required contour comes with choosing − sign, i.e $x = \frac{\mu f}{\mu+1}\left[1 - \sqrt{1-\left(\frac{\mu+1}{\mu-1}\right)\frac{y^2}{f^2}}\right]$



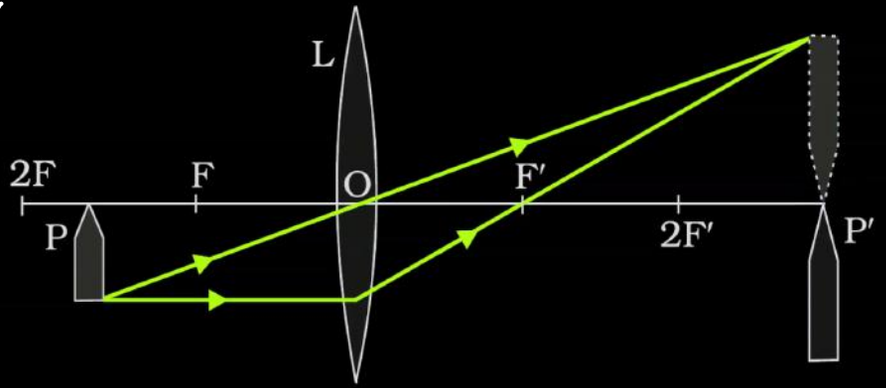
(This matches with Irodov's answer).

Experiment To Find Focal Length Of Convex Lens Using u-v Plot:



In this experiment, we mount a lens, object needle and image needle on an optical bench and plot the image vs object position wrt lens to get the focal length.

Consider the adjacent ray diagram, here, the tip P of the object needle forms an inverted image at P' which is located by positioning image needle to coincide with P' (we need to ensure that there is no parallax).



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PARALLAX ERROR: The image and image needle may appear to coincide from certain point of view due to both lying along same line of sight however they might not actually coincide. The error in readings due to this phenomenon is called parallax error. To remove parallax error, we must move head side to side while adjusting the position of image needle until the image actually coincides with image needle.

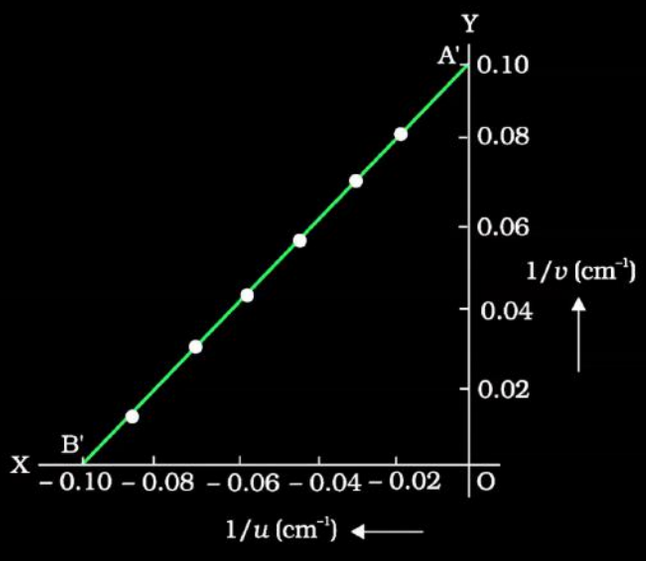
INDEX ERROR: We read the positions of lens, object needle and image needles from the location of their mounts, however this may lead to errors due to imperfect mounting or construction defect in the mount itself, the error introduced to this is called index error. To remove the index error, we take a stick of known length and place it between the lens and the object needle by adjusting the mounts until the stick is simultaneously touching the lens and the object needle. For this adjustment, the length of the stick represents true value of $|u|$ while difference in mount readings represents the measured value $|u_m|$, now index error is simply $|u_m| - |u|$, and index correction is then $|u| - |u_m|$. This correction must be added to the object distances to remove the index error. Similarly we can also remove the index error for the image pin.

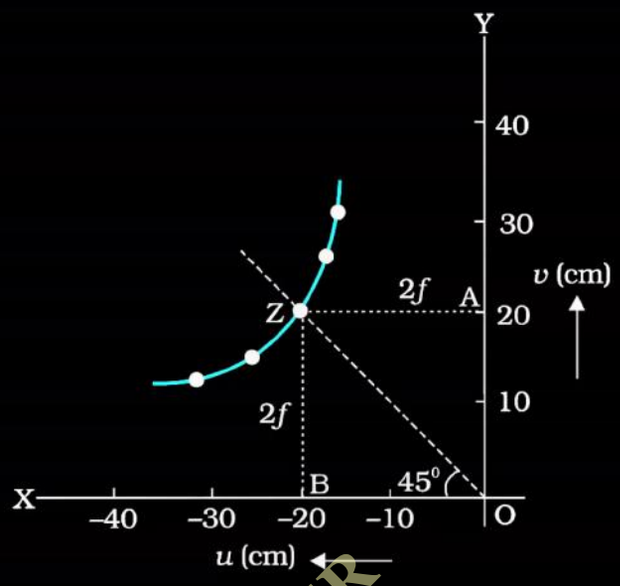
→ If the thickness of the lens is also taken in to account, index error is more correctly calculated as $|u_m| - (l + t/2)$ where l is the length of the stick and t is thickness of the lens. This is to account for the fact that optical centre is at the middle of the biconvex lens.

PLOTTING DATA POINTS AND FINDING FOCAL LENGTH:

Once we have taken several readings, we can plot either u-v graph or 1/u vs 1/v graph (see figures below).

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If we plot 1/u vs 1/v graph, focal length is given as

$$f = \frac{2}{OA' + OB'}$$ or if we plot u-v graph, we draw a bisector OZ and now $f = (OA + OB)/4$.

ERROR IN FOCAL LENGTH: We have $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ thus:

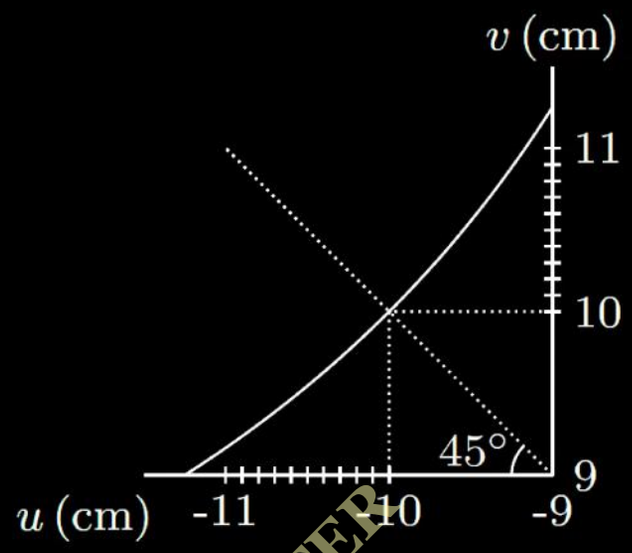
$\frac{\Delta f}{f^2} = \frac{\Delta v}{v^2} - \frac{\Delta u}{u^2}$, now as per NCERT, the error is found as $\boxed{\Delta f = f^2\left[\frac{\Delta u}{u^2} + \frac{\Delta v}{v^2}\right]}$ RIP correlated error!! ☺

(Actually errors in u and v are correlated because what we actually read are not direct values of u and v rather they are calculated as difference of coordinates, $x_2 - x_0$ and $x_1 - x_2$ where x_2, x_1 and x_0 are coordinates of lens, image and object pins.).

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Example : JEE 2006:

The graph between object distance u and the image distance v for a lens is as shown. What is the focal length of the lens with appropriate error limits.



Solution: Least count of the scale is 1mm. thus putting $u = -10$cm, $v = +10$ cm and $\Delta u = \Delta v = 0.1$ cm in formulae:

$\frac{\Delta f}{f^2} = \frac{\Delta v}{v^2} + \frac{\Delta u}{u^2}$ and $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$,

we get $f = 5.0$ cm and $\Delta f = 0.05$ cm thus $f = 5.0 \pm 0.05$ cm Ans.

Example : (JEE 2009):

A student performed the experiment for determination of focal length of a concave mirror by using u-v method using an optical bench of length 1.5 m. The focal length of the mirror used is 24 cm. The maximum error in the location of the image can be 0.2 cm. The 5 sets of (u,v) values recorded by the student (in cm) are: (42,56), (48,48), (60,40), (66,33) and (78,39). The data set(s) that cannot come from experiment and is (are) incorrectly recorded, is (are):

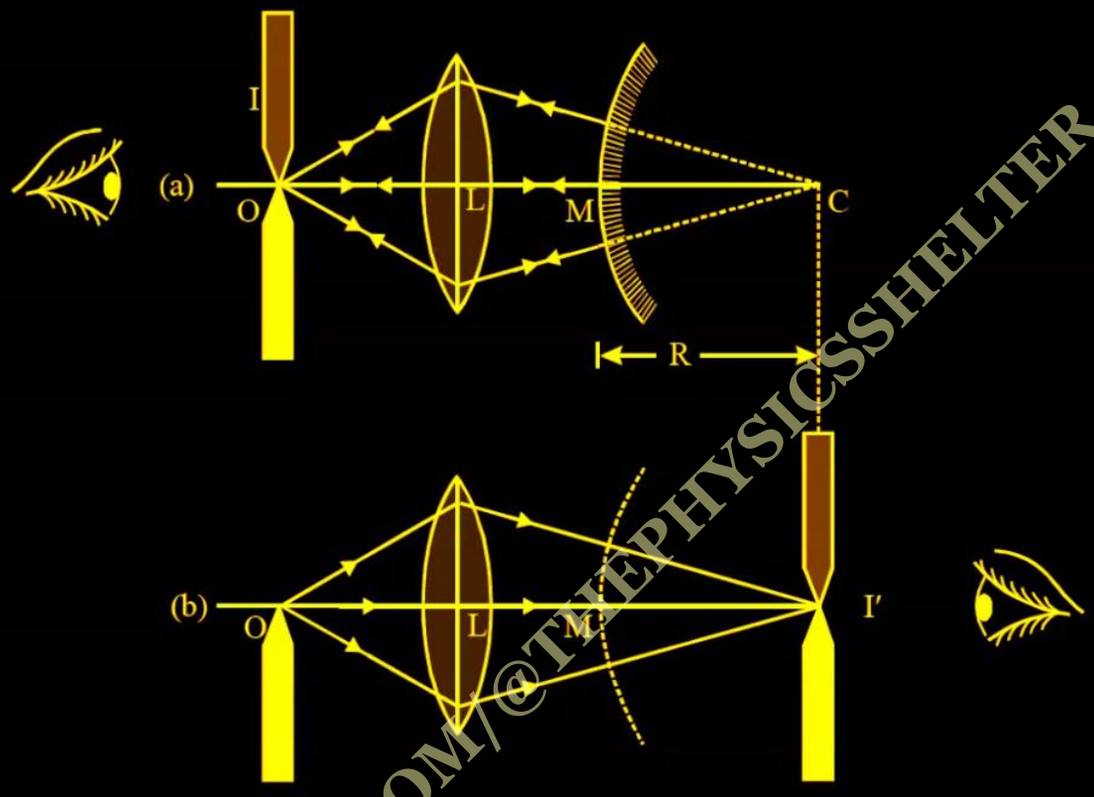
(A) (42,56) (B) (48,48) (C) (66,33) (D) (78,39)

Solution: Using mirror formula $1/v + 1/u = 1/f$, (u, v) pairs for given v value come as $(42, 56)$, $(48, 48)$, $(60, 40)$, $(66, 37.7)$ and $(78, 34.7)$, clearly the last two readings of v have a more than 0.2 cm difference from theoretical calculation, thus C and D are incorrectly recorded. C, D Ans.

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Experiment To Find Focal Length Of Convex Mirror Using Convex Lens:

Suppose a convex lens L is interposed between a convex mirror M and an object needle O as shown in Fig. (a). When the relative position of M, L and O are adjusted in such a way that there is no parallax between the object needle O and its image I, then in that position, the rays will fall normally on the convex mirror M. The rays which fall on the mirror normally should meet at the centre of curvature C of the mirror when produced (Fig. (a)) The distance MC gives the radius of curvature R. Half of the radius of curvature gives the focal length f of the mirror.



Focal length of the convex mirror using a convex lens

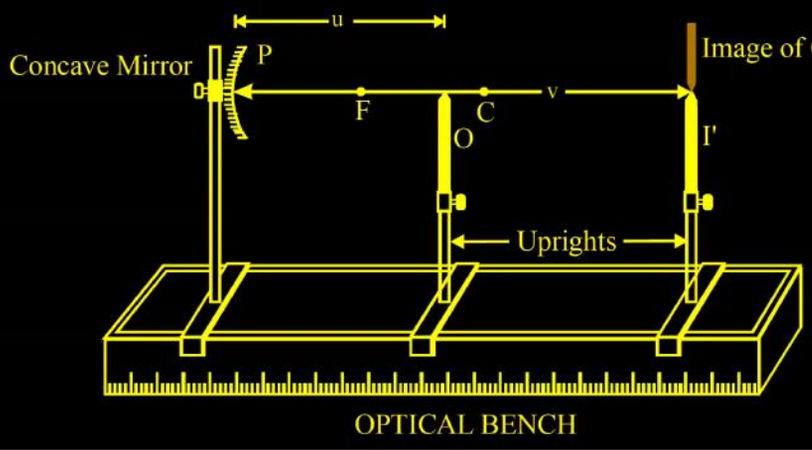
Now without disturbing the positions of the object O and the lens L, the convex mirror is removed and another needle is placed in the position of the image I′ of the object O, formed by the lens L by using parallax method as shown in Fig.

Measure MI′. Now, $f = \frac{R}{2} = \frac{MI'}{2}$

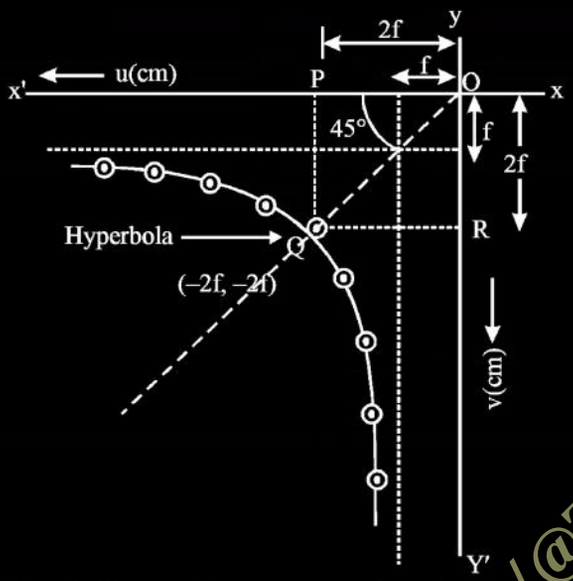
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Finding Focal Length Of Concave Mirror: The focal length of concave mirror can be calculated by a procedure similar to that for a convex lens, here we will mount the mirror and the two pins being on same side of the mirror. The set up for this is as shown in the following figure.

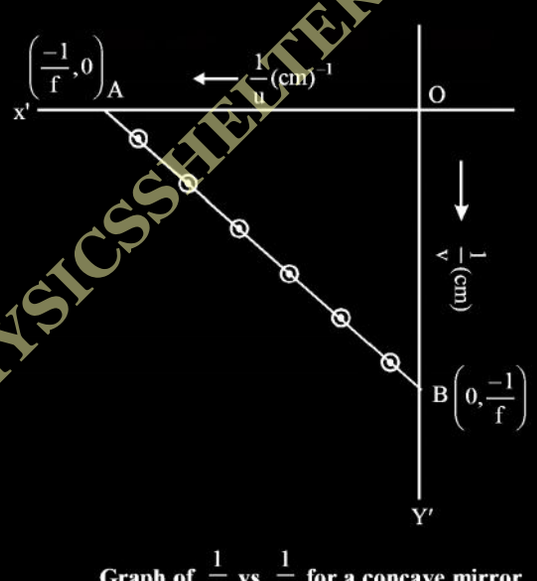
Here also we need to find the index errors for the object pin and the image pins. We collect the data for several values of u and then plot either u-v graph or 1/u vs 1/v graphs. These graphs are as shown:



OPTICAL BENCH



Graph of u vs. v for a concave mirror (rectangular hyperbola)



Graph of $\frac{1}{u}$ vs. $\frac{1}{v}$ for a concave mirror

Now from u-v graph, $f = -\frac{1}{4}[OP + OR]$ (OP & OA are magnitude)

from 1/v vs 1/u graph, $f = -\frac{1}{2}\left[\frac{1}{OA} + \frac{1}{OB}\right]$ (OA & OB are magnitudes).

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